

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396892
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	George; David et al.

External ear canal pressure regulation device

Abstract

An external ear canal pressure regulation device including a fluid flow generator, an earpiece having an axial earpiece bore, and a valved conduit fluidically coupled to the fluid flow generator and the axial earpiece bore, whereby the earpiece has a compliant earpiece external surface configured to sealably engage an external ear canal as a barrier between an external ear canal pressure and an ambient pressure.

Inventors: George; David (Scottsdale, AZ), Buckler; George (Phoenix, AZ), Crown; Timothy A. (Tempe, AZ), Sullivan; David Brice (Mechanicsburg, PA)

Applicant: NOCIRA, LLC (Tempe, AZ)

Family ID: 1000008782716

Assignee: NOCIRA, LLC (Tempe, AZ)

Appl. No.: 17/401733

Filed: August 13, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220202617 A1	Jun. 30, 2022

Related U.S. Application Data

continuation parent-doc US 14900607 US 11090194 WO PCT/US2014/044159 20140625 child-doc US 17401733
continuation-in-part parent-doc US 14292469 20140530 US 10251790 20190409 child-doc US 14900607
us-provisional-application US 61983865 20140424

Publication Classification

Int. Cl.: **A61F11/12** (20060101); **A61F11/00** (20220101); **A61F11/08** (20060101); **A61H9/00** (20060101); **A61H21/00** (20060101)

U.S. Cl.:

CPC **A61F11/12** (20130101); **A61F11/00** (20130101); **A61F11/08** (20130101); **A61H9/0007** (20130101); **A61H9/005** (20130101); **A61H9/0071** (20130101); **A61H21/00** (20130101); A61F11/085 (20220101); A61F2250/0013 (20130101); A61H2201/0153 (20130101); A61H2201/0207 (20130101); A61H2201/0214 (20130101); A61H2201/10 (20130101); A61H2201/105 (20130101); A61H2205/027 (20130101)

Field of Classification Search

CPC: A61F (11/00); A61F (11/12); A61F (2250/0069); A61H (9/0071); A61H (21/00); A61H (23/02); A61H (23/04); A61H (2201/0153); A61H (2201/1207); A61H (2201/123-1246); A61H (2201/5071); A61H (2205/027); A61H (9/0007); A61M (13/003); A61M (2205/3344); A61M (2205/362); A61M (2210/0662-0675); H04R (1/42)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
787443	12/1904	Godman et al.	N/A	N/A
841146	12/1906	Hasbrouck	N/A	N/A
853645	12/1906	Meyer	N/A	N/A
2176366	12/1938	Smith	N/A	N/A
2437490	12/1947	Watson et al.	N/A	N/A
2570675	12/1950	Morris	N/A	N/A
2652048	12/1952	Joers	N/A	N/A
3757769	12/1972	Arguimbau et al.	N/A	N/A
3872559	12/1974	Leight	N/A	N/A
4002161	12/1976	Klar et al.	N/A	N/A
4133984	12/1978	Watson et al.	N/A	N/A
4160449	12/1978	Wade	N/A	N/A
4206756	12/1979	Grossan	N/A	N/A
4244377	12/1980	Grams	N/A	N/A
4289143	12/1980	Canavesio et al.	N/A	N/A
4325386	12/1981	Katz	N/A	N/A
4349083	12/1981	Bennett	N/A	N/A
4472342	12/1983	Carr	N/A	N/A
4552137	12/1984	Strauss	N/A	N/A
4594058	12/1985	Fischell	N/A	N/A
4632104	12/1985	Conrow	N/A	N/A
4667676	12/1986	Guinta	N/A	N/A

4688582	12/1986	Heller et al.	N/A	N/A
4754748	12/1987	Antowski	N/A	N/A
4757807	12/1987	Densert et al.	N/A	N/A
4775370	12/1987	Berry	N/A	N/A
4809708	12/1988	Geisler et al.	N/A	N/A
4896380	12/1989	Kamitani	N/A	N/A
4896679	12/1989	St. Pierre	N/A	N/A
4964769	12/1989	Hass	N/A	N/A
4984579	12/1990	Burgert et al.	N/A	N/A
5024612	12/1990	van den Honert	604/36	A61F 11/10
5105822	12/1991	Stevens et al.	N/A	N/A
5131411	12/1991	Casali et al.	N/A	N/A
5228431	12/1992	Giarretto	N/A	N/A
5241967	12/1992	Yasushi et al.	N/A	N/A
5421818	12/1994	Arenberg	N/A	N/A
5431636	12/1994	Stangerup	N/A	N/A
5467784	12/1994	Mobley et al.	N/A	N/A
5476446	12/1994	Arenburg	N/A	N/A
5483027	12/1995	Krause	N/A	N/A
5483975	12/1995	Hirschebain	N/A	N/A
5488961	12/1995	Adams	N/A	N/A
5631965	12/1996	Chang et al.	N/A	N/A
5699809	12/1996	Combs et al.	N/A	N/A
5740258	12/1997	Goodwin-Johansson	N/A	N/A
5746725	12/1997	Shalon et al.	N/A	N/A
5755234	12/1997	Mobley et al.	N/A	N/A
5769891	12/1997	Clayton	N/A	N/A
5776179	12/1997	Ren et al.	N/A	N/A
5819745	12/1997	Mobley et al.	N/A	N/A
5865183	12/1998	Hirschebain	N/A	N/A
5868682	12/1998	Combe et al.	N/A	N/A
5944711	12/1998	Pender	N/A	N/A
6004274	12/1998	Nolan et al.	N/A	N/A
6016499	12/1999	Ferguson	N/A	N/A
6024726	12/1999	Hill	N/A	N/A
6129174	12/1999	Brown et al.	N/A	N/A
6139507	12/1999	Jeng	N/A	N/A
6159171	12/1999	Densert et al.	N/A	N/A
6186959	12/2000	Canfield et al.	N/A	N/A
6258067	12/2000	Hill	N/A	N/A
6296652	12/2000	Qingmin	N/A	N/A
6359993	12/2001	Birmhall	N/A	N/A
6430443	12/2001	Karell	N/A	N/A
6511437	12/2002	Nakamura et al.	N/A	N/A
6543445	12/2002	Hopper	N/A	N/A
6592512	12/2002	Stöckert et al.	N/A	N/A
6629938	12/2002	Engvall et al.	N/A	N/A
6725568	12/2003	Gronka	N/A	N/A
6748275	12/2003	Lattner et al.	N/A	N/A

6800062	12/2003	Epley	N/A	N/A
6820717	12/2003	Fleming et al.	N/A	N/A
6878128	12/2004	MacMahon et al.	N/A	N/A
6958043	12/2004	Hissong	N/A	N/A
6981569	12/2005	Stilp	N/A	N/A
7022090	12/2005	Engvall et al.	N/A	N/A
7162039	12/2006	Callahan	N/A	N/A
7179238	12/2006	Hissong	N/A	N/A
7189252	12/2006	Krueger	N/A	N/A
7268466	12/2006	Rasmussen	N/A	N/A
7352871	12/2007	Mozo	N/A	N/A
D570457	12/2007	Brown	N/A	N/A
7613519	12/2008	De Ridder	N/A	N/A
7766858	12/2009	Franz et al.	N/A	N/A
7779844	12/2009	Purcell et al.	N/A	N/A
7785346	12/2009	Blumberg	N/A	N/A
7797042	12/2009	Dietrich et al.	N/A	N/A
7833282	12/2009	Mandpe	N/A	N/A
7892180	12/2010	Epley	N/A	N/A
7959597	12/2010	Baker et al.	N/A	N/A
7988657	12/2010	Shapiro et al.	N/A	N/A
8020563	12/2010	Pfanstiehl	N/A	N/A
8047207	12/2010	Perez et al.	N/A	N/A
8052693	12/2010	Shahoian	N/A	N/A
8122892	12/2011	Johnson et al.	N/A	N/A
8142373	12/2011	Riles	N/A	N/A
8199919	12/2011	Goldstein et al.	N/A	N/A
8241224	12/2011	Keefe	N/A	N/A
8249285	12/2011	Killion et al.	N/A	N/A
8251925	12/2011	Keady et al.	N/A	N/A
8262717	12/2011	Rogers et al.	N/A	N/A
8267983	12/2011	Rogers et al.	N/A	N/A
8267984	12/2011	Rogers	N/A	N/A
8328830	12/2011	Pandit	N/A	N/A
8398562	12/2012	Keller	N/A	N/A
8414521	12/2012	Baker et al.	N/A	N/A
8442632	12/2012	Kullok et al.	N/A	N/A
8460356	12/2012	Rogers et al.	N/A	N/A
8506469	12/2012	Dietrich et al.	N/A	N/A
8515552	12/2012	Englehart	N/A	N/A
8550206	12/2012	Keady et al.	N/A	N/A
8568348	12/2012	Vlodaver	N/A	N/A
8603152	12/2012	Smith et al.	N/A	N/A
8625833	12/2013	Armwood	N/A	N/A
8666502	12/2013	Hartlep et al.	N/A	N/A
8688239	12/2013	Hartlep et al.	N/A	N/A
8696724	12/2013	Rogers	N/A	N/A
8858430	12/2013	Oyadiran et al.	N/A	N/A
8963914	12/2014	Rawat et al.	N/A	N/A
9039639	12/2014	George et al.	N/A	N/A

9168171	12/2014	Rogers	N/A	N/A
9186277	12/2014	George et al.	N/A	N/A
9283111	12/2015	Rogers et al.	N/A	N/A
9526653	12/2015	Rogers et al.	N/A	N/A
9532900	12/2016	Smith et al.	N/A	N/A
9579247	12/2016	Juto et al.	N/A	N/A
9655772	12/2016	Smith et al.	N/A	N/A
9744074	12/2016	Rogers	N/A	N/A
9849026	12/2016	Rogers et al.	N/A	N/A
10076464	12/2017	George et al.	N/A	N/A
10251790	12/2018	George et al.	N/A	N/A
10271992	12/2018	Hayahi et al.	N/A	N/A
10278868	12/2018	George et al.	N/A	N/A
10319475	12/2018	Croan et al.	N/A	N/A
10376695	12/2018	Ericco et al.	N/A	N/A
10760566	12/2019	George et al.	N/A	N/A
10772766	12/2019	Sullivan	N/A	N/A
11065444	12/2020	Ericco et al.	N/A	N/A
11090194	12/2020	George et al.	N/A	N/A
11096828	12/2020	George et al.	N/A	N/A
11246793	12/2021	George et al.	N/A	N/A
11859606	12/2023	George et al.	N/A	N/A
2001/0025191	12/2000	Montgomery	N/A	N/A
2002/0069883	12/2001	Hirchenbain	N/A	N/A
2003/0105450	12/2002	Dimick	N/A	N/A
2003/0195588	12/2002	Fischell et al.	N/A	N/A
2003/0220536	12/2002	Hissong	N/A	N/A
2003/0220585	12/2002	Hissong	N/A	N/A
2004/0097839	12/2003	Epley	N/A	N/A
2004/0163882	12/2003	Fleming et al.	N/A	N/A
2004/0225178	12/2003	Kriewall	N/A	N/A
2005/0065585	12/2004	Salas	N/A	N/A
2005/0165460	12/2004	Erfan	N/A	N/A
2005/0209516	12/2004	Fraden	N/A	N/A
2005/0267388	12/2004	Hanna	N/A	N/A
2006/0100681	12/2005	Salas Carpizo	N/A	N/A
2006/0197412	12/2005	Rasmussen	N/A	N/A
2006/0253087	12/2005	Vlodaver et al.	N/A	N/A
2006/0272650	12/2005	Hoogenakker et al.	N/A	N/A
2007/0040454	12/2006	Freudenberger et al.	N/A	N/A
2007/0112279	12/2006	Iseberg et al.	N/A	N/A
2007/0250119	12/2006	Tyler et al.	N/A	N/A
2007/0299362	12/2006	Epley et al.	N/A	N/A
2008/0011308	12/2007	Fleming	N/A	N/A
2008/0154183	12/2007	Baker et al.	N/A	N/A
2008/0168775	12/2007	Windheim et al.	N/A	N/A
2008/0208100	12/2007	Wolff	N/A	N/A
2008/0212787	12/2007	Goldstein et al.	N/A	N/A
2008/0220092	12/2007	Dipierro	N/A	N/A

2008/0240942	12/2007	Heinrich et al.	N/A	N/A
2008/0249439	12/2007	Tracey et al.	N/A	N/A
2008/0264464	12/2007	Lee et al.	N/A	N/A
2009/0012420	12/2008	Keller	N/A	N/A
2009/0082831	12/2008	Paul et al.	N/A	N/A
2009/0173353	12/2008	Pursell et al.	N/A	N/A
2009/0182399	12/2008	Sylvestre	N/A	N/A
2009/0228103	12/2008	Clayton	N/A	N/A
2009/0293886	12/2008	Dedrick et al.	N/A	N/A
2010/0002897	12/2009	Keady	N/A	N/A
2010/0030131	12/2009	Morris et al.	N/A	N/A
2010/0071707	12/2009	Wohl	N/A	N/A
2010/0071708	12/2009	Lenhardt	N/A	N/A
2010/0113991	12/2009	Wu	N/A	N/A
2010/0179490	12/2009	Connelly et al.	N/A	N/A
2010/0198282	12/2009	Rogers	N/A	N/A
2010/0211142	12/2009	Rogers	N/A	N/A
2010/0322454	12/2009	Ambrose et al.	N/A	N/A
2011/0079227	12/2010	Turcot et al.	N/A	N/A
2011/0097141	12/2010	Browm	N/A	N/A
2011/0098551	12/2010	Zhang	N/A	N/A
2011/0130786	12/2010	Clayton et al.	N/A	N/A
2011/0172739	12/2010	Mann et al.	N/A	N/A
2011/0184341	12/2010	Baker et al.	N/A	N/A
2011/0224493	12/2010	Oyadiran et al.	N/A	N/A
2011/0245902	12/2010	Katz	N/A	N/A
2011/0301572	12/2010	Vlodaver et al.	N/A	N/A
2011/0313481	12/2010	De Vos	607/42	A61N 1/0546
2012/0046607	12/2011	Syk	N/A	N/A
2012/0203309	12/2011	Englehart	N/A	N/A
2012/0265093	12/2011	Allen et al.	N/A	N/A
2012/0296268	12/2011	Vlodavaer et al.	N/A	N/A
2012/0302859	12/2011	Keefe	N/A	N/A
2012/0310077	12/2011	Rogers	N/A	N/A
2012/0310313	12/2011	Rogers et al.	N/A	N/A
2012/0318605	12/2011	Brown	N/A	N/A
2013/0123889	12/2012	Katz et al.	N/A	N/A
2013/0136285	12/2012	Naumann	N/A	N/A
2013/0152949	12/2012	Simon	N/A	N/A
2013/0177179	12/2012	Ambrose et al.	N/A	N/A
2013/0183173	12/2012	Kohli et al.	N/A	N/A
2013/0282070	12/2012	Cowan et al.	N/A	N/A
2013/0303953	12/2012	Lattner	N/A	N/A
2013/0304103	12/2012	Burres	N/A	N/A
2013/0310907	12/2012	Rogers et al.	N/A	N/A
2013/0324932	12/2012	Cogley	N/A	N/A
2013/0331823	12/2012	Askem et al.	N/A	N/A
2014/0069442	12/2013	Lewis et al.	N/A	N/A
2014/0088671	12/2013	Rogers et al.	N/A	N/A

2014/0243941	12/2013	Rogers et al.	N/A	N/A
2014/0249608	12/2013	Rogers	N/A	N/A
2014/0275827	12/2013	Gill et al.	N/A	N/A
2014/0309718	12/2013	Smith et al.	N/A	N/A
2014/0334652	12/2013	Gebert	N/A	N/A
2014/0344740	12/2013	Kaula et al.	N/A	N/A
2014/0365175	12/2013	Packer et al.	N/A	N/A
2015/0005661	12/2014	Trammell	N/A	N/A
2015/0141879	12/2014	Harper et al.	N/A	N/A
2015/0320591	12/2014	Smith et al.	N/A	N/A
2015/0320592	12/2014	Black et al.	N/A	N/A
2015/0324544	12/2014	Maslowski et al.	N/A	N/A
2015/0328413	12/2014	Cain	N/A	N/A
2015/0335466	12/2014	Schöggler	N/A	N/A
2015/0374538	12/2014	Rogers	N/A	N/A
2016/0067099	12/2015	Hayashi	N/A	N/A
2016/0166203	12/2015	Goldstein	N/A	N/A
2016/0279435	12/2015	Hyde et al.	N/A	N/A
2016/0346117	12/2015	Rogers et al.	N/A	N/A
2016/0378945	12/2015	Mian et al.	N/A	N/A
2017/0105876	12/2016	O'Connell, Sr. et al.	N/A	N/A
2017/0109988	12/2016	O'Connell, Sr. et al.	N/A	N/A
2017/0135854	12/2016	Rogers et al.	N/A	N/A
2017/0235889	12/2016	Main et al.	N/A	N/A
2018/0008457	12/2017	Smith et al.	N/A	N/A
2018/0106244	12/2017	Wang et al.	N/A	N/A
2018/0125748	12/2017	Goldenberg et al.	N/A	N/A
2020/0121544	12/2019	George et al.	N/A	N/A
2021/0222684	12/2020	George et al.	N/A	N/A
2021/0228414	12/2020	George et al.	N/A	N/A
2021/0330928	12/2020	George et al.	N/A	N/A
2022/0226158	12/2021	George et al.	N/A	N/A
2022/0370286	12/2021	George et al.	N/A	N/A
2024/0271609	12/2023	George	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1136751	12/1981	CA	N/A
1222464	12/1986	CA	N/A
1241152	12/1987	CA	N/A
2003452	12/1989	CA	N/A
2275057	12/1998	CA	N/A
2 337 076	12/1999	CA	N/A
2429560	12/2003	CA	N/A
2075517	12/1990	CN	N/A
2418864	12/2000	CN	N/A
1308513	12/2000	CN	N/A
2530645	12/2002	CN	N/A
2721057	12/2004	CN	N/A

1791370	12/2005	CN	N/A
2912525	12/2006	CN	N/A
200945215	12/2006	CN	N/A
201143258	12/2007	CN	N/A
201164541	12/2007	CN	N/A
101668497	12/2009	CN	N/A
201505220	12/2009	CN	N/A
201524178	12/2009	CN	N/A
201558360	12/2009	CN	N/A
201870809	12/2010	CN	N/A
202036187	12/2010	CN	N/A
202185057	12/2011	CN	N/A
102484761	12/2011	CN	N/A
102551957	12/2011	CN	N/A
202313927	12/2011	CN	N/A
102647966	12/2011	CN	N/A
202477966	12/2011	CN	N/A
202505833	12/2011	CN	N/A
102892392	12/2012	CN	N/A
102986250	12/2012	CN	N/A
37 83 917	12/1992	DE	N/A
102011008802	12/2011	DE	N/A
0 026 247	12/1980	EP	N/A
0 400 900	12/1989	EP	N/A
1 027 863	12/1999	EP	N/A
2 207 366	12/2009	EP	N/A
2 990 017	12/2015	EP	N/A
2 605 516	12/1987	FR	N/A
2 779 944	12/1998	FR	N/A
1432572	12/1975	GB	N/A
1522031	12/1977	GB	N/A
2054387	12/1980	GB	N/A
2185688	12/1986	GB	N/A
2343263	12/1999	GB	N/A
2479891	12/2010	GB	N/A
1214840	12/1989	IT	N/A
S 57-188245	12/1981	JP	N/A
H 02-220650	12/1989	JP	N/A
H 07-111987	12/1994	JP	N/A
H 11-514898	12/1998	JP	N/A
2002-519150	12/2001	JP	N/A
2003-018359	12/2002	JP	N/A
2006-345903	12/2005	JP	N/A
2009-022699	12/2008	JP	N/A
2010-233643	12/2009	JP	N/A
2010-535542	12/2009	JP	N/A
2011-217986	12/2010	JP	N/A
2013-068448	12/2012	JP	N/A
2013-102784	12/2012	JP	N/A
2020-44371	12/2019	JP	N/A

10-1273296	12/2012	KR	N/A
PA03005598	12/2003	MX	N/A
2010014470	12/2010	MX	N/A
2011006854	12/2010	MX	N/A
2012007726	12/2011	MX	N/A
90 333	12/2009	RU	N/A
WO 1986/01399	12/1985	WO	N/A
WO 1994/22372	12/1993	WO	N/A
WO 1996/23293	12/1995	WO	N/A
WO 1997/23178	12/1996	WO	N/A
WO 2000/001331	12/1999	WO	N/A
WO 2000/001346	12/1999	WO	N/A
WO 2000/010484	12/1999	WO	N/A
WO 2000/010627	12/1999	WO	N/A
WO 2000/010848	12/1999	WO	N/A
WO 2001/19244	12/2000	WO	N/A
WO 2003/075761	12/2002	WO	N/A
WO 2004/064672	12/2003	WO	N/A
WO 2004/100844	12/2003	WO	N/A
WO 2006/003910	12/2005	WO	N/A
WO 2006/009545	12/2005	WO	N/A
WO 2007/084674	12/2006	WO	N/A
WO 2007/118092	12/2006	WO	N/A
WO 2007/145853	12/2006	WO	N/A
WO 2008/036368	12/2007	WO	N/A
WO 2008/064230	12/2007	WO	N/A
WO 2008/086187	12/2007	WO	N/A
WO 2008/128173	12/2007	WO	N/A
WO 2008/153588	12/2007	WO	N/A
WO 2009/020862	12/2008	WO	N/A
WO 2009/050306	12/2008	WO	N/A
WO 2009/077902	12/2008	WO	N/A
WO 2010/005899	12/2009	WO	N/A
WO 2010/016925	12/2009	WO	N/A
WO 2010/085196	12/2009	WO	N/A
WO 2011/075573	12/2010	WO	N/A
WO 2011/075574	12/2010	WO	N/A
WO 2012/007193	12/2011	WO	N/A
WO 2012/083098	12/2011	WO	N/A
WO 2012/083102	12/2011	WO	N/A
WO 2012/083106	12/2011	WO	N/A
WO 2012/083126	12/2011	WO	N/A
WO 2012/083151	12/2011	WO	N/A
WO 2013/075255	12/2012	WO	N/A
WO 2014/120947	12/2013	WO	N/A
WO 2014/175257	12/2013	WO	N/A
WO 2014/210457	12/2013	WO	N/A
WO 2015/009421	12/2014	WO	N/A
WO 2015/074060	12/2014	WO	N/A
WO 2016/022761	12/2015	WO	N/A

WO 2017/040739	12/2016	WO	N/A
WO 2017/040741	12/2016	WO	N/A
WO 2017/040747	12/2016	WO	N/A
WO 2017/079783	12/2016	WO	N/A
WO 2017/197150	12/2016	WO	N/A
WO 2018/106839	12/2017	WO	N/A
WO 2018/157143	12/2017	WO	N/A
WO 2019/246456	12/2018	WO	N/A

OTHER PUBLICATIONS

Akerman, et al. Pearls and pitfalls in experimental in vivo models of migraine: Dural trigeminovascular nociception. *Cephalalgia*, 2013, 33 (8), pp. 557-592. cited by applicant

Baguley et al. Does caloric vestibular stimulation modulate tinnitus? *Neuroscience Letters*, Mar. 2011, 492(1), pp. 52-54. cited by applicant

Baier, et al.: "Vestibular-Evoked Myogenic Potentials in "Vestibular Migraine" and Meniere's Disease," *Ann. N.Y. Acad. Sci.*, May 2009, 1164, pp. 324-327. cited by applicant

Becker: Weather and migraine: Can so many patients be wrong? *Cephalalgia*, Mar. 2011, 31(4), pp. 387-390. cited by applicant

Berthold Langguth, Verena Hund, Volker Busch, et al., "Tinnitus and Headache," *BioMed Research International*, vol. 2015, Article ID 797416, 7 pages, 2015. <https://doi.org/10.1155/2015/797416> (Year: 2015) in 7 pages. cited by applicant

Bolay et al.: "Does Low Atmospheric Pressure Independently Trigger Migraine?" *Headache*, Oct. 2011, 51(9), pp. 1426-1430. cited by applicant

Breathometer. Breathometer—The World's First Smartphone Breathalyzer. Website, <http://www.breathometer.com>, originally downloaded Jun. 19, 2014, 8 total pages. cited by applicant

Cadwell. Sierra Wave. Website, <http://www.cadwell.com>, originally downloaded Feb. 27, 2014, 1 page. cited by applicant

Cathcart, et al., "Pain sensitivity mediates the relationship between stress and headache intensity in chronic tension-type headache", Nov. 2012 (Year: 2012) in 5 pages. cited by applicant

Cranial Nerves—Wikipedia, axes, printed Aug. 16, 2019 in 12 pages. cited by applicant

Croley, Christen, "Mechanicsburg doctor develops new migraine therapy," *The Sentinel*, Nov. 9. 2012. cited by applicant

DaSilva, et al.: "tdCS-Induced Analgesia and Electrical Fields in Pain-Related Neural Networks in Chronic Migraine," *The Journal of Head and Face Pain*, Sep. 2012, 52, pp. 1283-1295. cited by applicant

Dirckx et al. Human tympanic membrane deformation under static pressure. *Hearing Research*, Jan. 1991, 51(1), pp. 93-106. cited by applicant

Doherty, Colleen. "The Link Between Migraines and Tinnitus". *Verywell Health*, Nov. 23, 2019, <https://www.verywellhealth.com/link-between-migraines-and-tinnitus-4077631#citation-10> (Year: 2019) in 4 pages. cited by applicant

Facebook. Zōk: The first migraine and headache solution, Webpage, <https://www.facebook.com>, originally downloaded May 18, 2017, 10 pages total. cited by applicant

Fasold et al. Human Vestibular Cortex as Identified with Caloric Stimulation in Functional Magnetic Resonance Imaging. *Neuroimage*, Nov. 2002, 17(3), pp. 1384-1393. cited by applicant

Ferrotec. Thermal Solutions. Website: <http://thermal.ferrotec.com>, originally downloaded Feb. 27, 2014, 1 page. cited by applicant

Ferrotec. Thermoelectric Technical Reference—Installation of Thermoelectric Modules. Website, <http://thermal.ferrotec.com>, originally downloaded May 21, 2014, 4 total pages. cited by applicant

Ferrotec. Thermoelectric Technical Reference—Introduction to Thermoelectric Cooling. Website,

<http://forrotec.com>, originally downloaded Feb. 27, 2014, 2 total pages. cited by applicant

Frangos E, Ellrich J, Komisaruk B. Non-invasive access to the vagus nerve central projections via electrical stimulation of the external ear: fMRI evidence in humans. *Brain Stimul.* Dec. 6, 2014. 8(3), 624-636 in 13 pages. cited by applicant

George et al. Safety and usability factors in development of a novel, automated treatment device for acute migraine. *Biomedical sciences instrumentation. Biomedical sciences instrumentation*, Jan. 2017, 53, pp. 398-403. cited by applicant

Hahn: "Let Me Blow in Your Ear, for Migraine Treatment, of Course," *Smile Columbia Dentistry*, <https://www.tmjttreatments.com>, originally downloaded Apr. 25, 2016, 2 pages total. cited by applicant

Hu et al. Burden of migraine in the United States: disability and economic costs. *Arch. Intern. Med.*, Apr. 1999, 159, pp. 813-818. cited by applicant

Janetta Neurovascular Compress in Cranial Nerve and Systemic Disease. *Ann Surg*, Oct. 1980, 192 (4), pp. 518-524. cited by applicant

Job et al. Cortical Representation of Tympanic Membrane Movements due to Pressure Variation: An $\pm$ MRI Study Human Brain Mapping, May 2011, 32(5), pp. 744-749. cited by applicant

Kanzara T, Hall A, Virk J, Leung B, Singh A. Clinical anatomy of the tympanic nerve: A review. *World J Otorhinolaryngol.* Nov. 2014; 4(4), 17-22 in 8 pages. cited by applicant

Kickstarter. Zōk: The first headache product that solves migraines and headaches. Website, <https://www.funded.today>, originally downloaded May 18, 2017, 3 pages total. cited by applicant

Kiyokawa J., Yamaguchi K, Okada R, Maehara T, Akita K. Origin, course and distribution of the nerves to the posterosuperior wall of the external acoustic meatus. *Anat Sci Int.* Mar. 2014; 89(4), 238-245. cited by applicant

Klingner et al.: "Components of vestibular cortical function," *Behavioral Brain Research*, Jan. 2013, 236(1), pp. 194-199. cited by applicant

Kolev. How caloric vestibular irrigation influences migraine attacks. *Cephalalgia.* Aug. 1990, vol. 10 issue 4, pp. 167-169 (abstract only). cited by applicant

Lifting the Burden. The Global Campaign Against Headache. Website, <http://www.I-t-b.org>, originally downloaded Feb. 27, 2014, 1 page. cited by applicant

Liszewski: Ear Pressure Equalizer. Website, <http://www.ohgizmo.com>, originally downloaded Dec. 18, 2013, 1 page. cited by applicant

Long Island news12.com. Long Island Naturally: Migraines. Website video, <http://longisland.news12.com/multimedia/long-island-naturally-migraines-1.6501113>, Nov. 26, 2013, 3 total pages. cited by applicant

Mayr: *The Origins of Feedback Control*. M.I.T. Press, 1970. cited by applicant

McGeoch et al. Vestibular stimulation can relieve central pain of spinal origin. *Spinal Cord*, Nov. 2008, 46(11), pp. 756-757. cited by applicant

Medscape. Peripheral Nerve Stimulator-Train of Four Monitoring. Website, <http://emedicine.medscape.com>, originally downloaded Feb. 27, 2014, 2 total pages. cited by applicant

Medtronic. Meniett Device for Meniere's Disease. Meniett Low-Pressure Pulse Generator device. Website, <http://www.medtronic.com>, originally downloaded Feb. 27, 2014, 2 total pages. cited by applicant

Medtronic. Restore Life's Balance with Meniett Therapy. The Meniett Device for Meniere's Disease. On-line article, <http://www.medtronic.com>, originally downloaded Mar. 13, 2015, 2 total pages. cited by applicant

Meng et al. Migraine Prevention with a Supraorbital Transcutaneous Stimulator: A Randomized Controlled Trial. *Neurology*, Sep. 2013, 81, pp. 1102-1103. cited by applicant

Minen. Tinnitus and Headache. American Migraine Foundation, website, downloaded Feb. 8, 2017, 3 pages total. cited by applicant

Mosquera et al. Vagus Nerve Stimulation in Patients with Migraine. *Rev Neurol*, 2013, 57(2), English Abstract. cited by applicant

Nagai et al. Encapsulated nerve corpuscles in the human tympanic membrane. *Archives of Otorhinolaryngology*, 1989, 246(3), pp. 169-172. cited by applicant

New York Health Solutions. Migraine Headaches. Website, <http://www.nyhealthsolutions.com>, originally downloaded May 23, 2014, 2 pages. cited by applicant

Nihashi et al. Representation of the ear in human primary somatosensory cortex. *Neuroimage*, Feb. 2001, 13(2), pp. 295-304 (abstract only). cited by applicant

Olesen et al. Emerging Migraine treatments and drug targets. *Trends in Pharmacological Sciences*, 2011, 32(6), pp. 352-359. cited by applicant

Pederson et al. Neurostimulation in cluster headache: A review of current progress. *Cephalalgia*, 2013, 33(14), pp. 1179-1193. cited by applicant

Pietrobon, Migraine: new molecular mechanism. *Neuroscientist*. Aug. 2005, vol. 11, Issue 4, pp. 373-386 (abstract only). cited by applicant

Porta-Etessam et al. Neuro-otological symptoms in patients with migraine. *Neurologia*, Mar. 2011, 26(2), pp. 100-104. cited by applicant

Ramachandran et al. Rapid Relief of Thalamic Pain Syndrome Induced by Vestibular Caloric Stimulation. *Neurocase*, Jun. 2007, 13(3), pp. 185-188. cited by applicant

Sakata et al. Air pressure-sensing ability of the middle ear—Investigation of sensing regions and appropriate measurement conditions. *Auris Nasus Larynx*, Aug. 2009, 36(4), pp. 393-399. cited by applicant

Sameiro-Barbosa et al. Sensory Entrainment Mechanisms in Auditory Perception: Neural Synchronization Cortico-Striatal Activation. *Frontiers in Neuroscience*, Aug. 2016, vol. 10, Article 361, 8 pages. cited by applicant

Saunders R, Tympanic membrane sensation. *Brain*. 1985, 108, 378-404 in 18 pages. cited by applicant

Schoenen et al. Migraine prevention with a supraorbital transcutaneous stimulator. *Neurology*, 2013, 80(8), pp. 697-704. cited by applicant

Schulman. Breath-Holding, Head Pressure, and Hot Water: An Effective Treatment for Migraine Headache. *Headache*, Nov.-Dec. 2002, 42(10), pp. 1048-1050. cited by applicant

Scion Neurostim. Therapeutic Neuromodulation via Caloric Vestibular Stimulation. Thermoneuromodulation (TNM). Slides for presentation, dated Sep. 2015, 12 pages total. cited by applicant

Sheftell, F, Steiner, TJ, Thhomas, H. Harry Potter and the Curse of Headache. *Headache: The Journal of Head and Face Pain*. Jun. 2007, vol. 47, Issue 6, pp. 911-916 (abstract only) in 1 page. cited by applicant

Shevel, “Headaches and tinnitus: correlation found”, May 2008 (Year: 2006). cited by applicant

Silberstein et al.: “Botulinum Toxin Type A as a Migraine Preventive Treatment,” *The Journal of Head and Face Pain*, Jun. 2000, 40, pp. 445-450. cited by applicant

Smartproducts. Series 100—Cartridge Specialty Check Valves and Pressure Relief Valves. Online catalog, www.smartproducts.com, originally downloaded Mar. 28, 2014, 2 total. cited by applicant

Stender, DR., “Easing Migraine Symptoms with a Simple Puff of Air into the Ear,” *Pasadena Pain Management*, <http://www.pasadenapainmanagement.com>, downloaded Apr. 25, 2016, 5 pages total. cited by applicant

Stovnver, LJ, et al. The global burden of headache: a documentation of headache prevalence and disability worldwide. *Cephalalgia*, 2007. vol. 27, pp. 193-210. cited by applicant

Sullivan: “Ear Insufflation as a Novel Therapy Which Produces Rapid Relief of Migraine Headache—a Case Study,” *Funct Neurol Rehabil Egon* 2013; vol. 3, Issue 1, pp. 93-107. Published on Jun. 7, 2013. Received on Jan. 2, 2013. Revised Jan. 28, 2013. Accepted Feb. 15, 2013. cited by applicant

Sullivan: "Ear Insufflation Produces Rapid and Significant Relief of Trigeminal Neuralgia," *Funct Neurol Rehabil* Egon 2013; vol. 3, Issue 4, pp. 1-6. Published on May 26, 2014. Received on Jun. 21, 2013. Revised Dec. 24, 2013. Accepted Jan. 12, 2014. cited by applicant

Tekdemir I, Aslan A, Elhan A., A clinico-anatomic study of the auricular branch of the vagus nerve and Arnold's ear-cough reflex. *Surg Radiol Anat*. 1998. 20(4), 253-257 in 5 pages. cited by applicant

Tekdemir I, Aslan A, Tuccar E, He C, Elhan A, Deda H. An anatomical study of the tympanic branch of the glossopharyngeal nerve (nerve of Jacobson). *Ann Anat*. Aug. 1998; 180(4): 349-52 in 4 pages. cited by applicant

Transcript of News Story, Aug. 22, 2013, video available at: <https://www.facebook.com/178787878873891/videos/10201196245541704/>. cited by applicant

"New Migraine Therapy," Aug. 22, 2013, video available at <https://www.facebook.com/178787878873891/videos/10201196245541704/>. cited by applicant

Transcript of News Story, Nov. 13, 2013, video available at: <https://www.facebook.com/178787878873891/videos/treatment-for-migraines-and-trigeminal-neuralgia/10201781732138503/>. cited by applicant

"Revolutionary Pain Therapy," Nov. 13, 2013, video available at <https://www.facebook.com/178787878873891/videos/treatment-for-migraines-and-trigeminal-neuralgia/10201781732138503/>. cited by applicant

Transcript of News Story, Jul. 7, 2014, video available at: <https://www.facebook.com/178787878873891/videos/681870651898942/>. cited by applicant

"New Therapy for Migraines," Jul. 7, 2014, video available at <https://www.facebook.com/178787878873891/videos/treatment-for-migraines-and-trigeminal-neuralgia/10201781732138503/>. cited by applicant

Transcript of Webinar, Apr. 10, 2013, video available at: https://www.anymeeting.com/WebConference/RecordingDefault.aspx?c_psrId=ED57DC868548. cited by applicant

A novel application to resolve migraine headaches—A Functional video available at: https://www.anymeeting.com/WebConference/RecordingDefault.aspx?c_psrId=ED57DC868548. cited by applicant

Ultimate Ears. Ultimate Ears Custom In-Ear Monitors. Website, <http://pro.ultimateears.com>, originally downloaded Feb. 27, 2014, 3 total pages. cited by applicant

Von Korff, et al., "Assessing headaches severity. *New Directions*", Jul. 1994 (Year: 1994). cited by applicant

Westone. Occupational Earpieces. Website, <http://www.westone.com>, originally downloaded Feb. 27, 2014, 2 total pages. cited by applicant

Widemar L, Hellstrom S, Schultzberg M, Stenfors LE. Autonomic innervation of the tympanic membrane. An immunocytochemical and histofluorescence study. *Acta Otolaryngol*. Jul.-Aug. 1985;100(1-2):58:65 in 9 pages. cited by applicant

Wikipedia. Microcurrent electrical neuromuscular stimulator. Website, <http://en.wikipedia.org>, originally downloaded Feb. 27, 2014, 3 total pages. cited by applicant

Wikipedia. Somatosensory evoked potential. Website, <http://en.wikipedia.org>, originally downloaded Feb. 27, 2014, 5 pages total. cited by applicant

Wikipedia. Transcutaneous electrical nerve stimulation. Website, <http://en.wikipedia.org>, originally downloaded Feb. 27, 2014, 5 pages total. cited by applicant

World Health Organization. Headache disorders. Website, <http://www.who.int>, originally downloaded Feb. 27, 2014, 4 total pages. cited by applicant

Chinese Patent Application No. 201480042665.7; Office Action dated Jan. 22, 2017, 26 pages total. cited by applicant

Chinese Office Action, dated Sep. 4, 2017, re CN Application No. 201480042665.7, 6 pages total.

cited by applicant
European Supplemental Search Report dated Dec. 8, 2016, re EP Application No. 14826160.5, 8 pages total. cited by applicant
European Office Action, dated Nov. 24, 2017, re EP Application No. 14816984.0, 6 pages total. cited by applicant
International Search Report and Written Opinion in co-pending application No. PCT/US2018/019981, mailed Jun. 27, 2018 in 15 pages. cited by applicant
New Zealand Patent Application No. 713887; Office Action mailed Feb. 20, 2017, 9 pages total. cited by applicant
PCT International Patent Cooperation Treaty Patent Application No. PCT/US14/044159, filed Jun. 25, 2014. cited by applicant
PCT International Patent Application No. PCT/US2014/0066191; Written Opinion of the International Searching Authority mailed Feb. 26, 2015, 7 pages total. cited by applicant
U.S. Appl. No. 61/841,111, filed Jun. 28, 2013. cited by applicant
U.S. Appl. No. 14/292,469, filed May 30, 2014. cited by applicant
U.S. Appl. No. 61/863,317, filed Aug. 7, 2013. cited by applicant
U.S. Appl. No. 61/905,616, filed Nov. 18, 2013. cited by applicant
U.S. Appl. No. 61/983,865, filed Apr. 24, 2014. cited by applicant
U.S. Appl. No. 07/286,744, filed Dec. 19, 1988. cited by applicant
Teixido, Michael: "Migraine—More than a Headache," Dec. 15, 1999, ENT and Allergy of Delaware (Year: 1999). cited by applicant
U.S. Appl. No. 18/949,671, filed Nov. 15, 2024. cited by applicant

Primary Examiner: Sippel; Rachel T

Attorney, Agent or Firm: Knobbe, Martens, Olson & Bear, LLP

Background/Summary

(1) This application is a continuation of U.S. Non-Provisional patent application Ser. No. 14/900,607, filed Dec. 21, 2015, now U.S. Pat. No. 11,090,194, issued Aug. 17, 2021, which is a national phase application of International Patent Cooperation Treaty Patent Application No. PCT/US2014/044159, filed on Jun. 25, 2014, which is a continuation-in-part of U.S. Non-Provisional patent application Ser. No. 14/292,469, filed May 30, 2014, now U.S. Pat. No. 10,251,790, issued Apr. 9, 2019, and claims the benefit of U.S. Provisional Patent Application No. 61/983,865, filed Apr. 24, 2014, U.S. Provisional Patent Application No. 61/863,317, filed Aug. 7, 2013, and U.S. Provisional Patent Application No. 61/841,111, filed Jun. 28, 2013, each of which is hereby incorporated by reference herein.

I. BACKGROUND OF THE INVENTION

(1) Pain or discomfort associated with a disorder, including neurologically-mediated disorders such as craniofacial pain syndromes or headache syndromes, may negatively impact the quality of life of the sufferer. In addition to the burden upon the individual, chronic neurological conditions may be a significant strain upon family members, employers, and the healthcare system.

(2) Regarding migraine headaches, concomitant symptoms such as pain, nausea, aura, photophobia, dysesthesias, dizziness, vertigo, and dysequilibrium may represent a significant burden to the population. Epidemiological studies indicate that, in the United States, approximately 18% of women and 6% of men experience frequent migraine headaches and 2% of the general population suffer from chronic migraine headaches. Additionally, persons suffering with chronic migraine

headaches or other headaches of similar severity and disability may be at a significantly greater risk for depression and attempted suicide. Thus, it is prudent for clinicians and researchers to continue searching for effective devices and methods to alleviate the symptoms associated with these disorders or to treat the disorders.

(3) Standard pharmaceutical therapies for migraine headaches may generally be prescribed to prevent pain or to relieve pain. The various agents which fall under these two broad categories may exhibit a wide range of effectiveness and also incur varying degrees of side effects. From the perspective of economics, the expense of these medications may be a major source of financial burden on the consumer. Moreover, advanced interventions such as botulinum toxin injections, nerve blockades, neurosurgical alterations, and implanted electrical stimulators may significantly increase costs associated with treatment, while subjecting patients to potential changes in their anatomy and physiology, with no guarantee of complete or permanent symptomatic relief or disorder resolution.

(4) There is a burgeoning field of understanding and applications within the neurosciences which seek to affect positive physiological changes in the nervous system through non-pharmaceutical and non-surgical applications. This field of 'functional neurology' views the human nervous system as a receptor driven system, which may be activated and stimulated in specific ways to produce adaptive, long-term changes through the process of neuroplasticity. This approach to neurorehabilitation utilizes, but not necessarily exclusively includes, various forms and patterns of receptor activation or deactivation to promote positive neurophysiological adaptations within the central nervous system, including the brain, brainstem, and spinal cord, which may promote physiological function of associated tissues, organs, and systems.

(5) There would be a substantial advantage in providing a device or methods which can generate one or more stimuli which can alleviate one or more symptoms associated with a disorder, such as craniofacial pain syndromes or headache syndromes, or treat one or more disorders.

II. DISCLOSURE OF THE INVENTION

(6) A broad object of particular embodiments of the invention can be to provide an external ear canal pressure regulation device including a fluid flow generator which generates a fluid flow; a valved conduit fluidically coupled to the fluid flow generator, the valved conduit having a first fluid flow conduit interruptible by one or more valves to unidirectionally regulate the fluid flow in the first fluid flow conduit; and an earpiece having an axial earpiece bore which communicates between an earpiece first end and an earpiece second end, the axial earpiece bore fluidically coupled to the valved conduit opposite the fluid flow generator, the earpiece having a compliant earpiece external surface configured to sealably engage an external ear canal as a barrier between an external ear canal pressure and an ambient pressure.

(7) Another broad object of particular embodiments of the invention can be to provide an external ear canal pressure regulation device having the valved conduit coupled in a first configuration with the fluid flow generator and the earpiece to unidirectionally regulate the fluid flow in a first direction in the first fluid flow conduit such that the fluid flow can egress from the axial earpiece bore of the earpiece toward the external ear canal, thereby achieving an external ear canal pressure greater than the ambient pressure.

(8) Another broad object of particular embodiments of the invention can be to provide an external ear canal pressure regulation device having the valved conduit coupled in a second configuration with the fluid flow generator and the earpiece to unidirectionally regulate the fluid flow in a second direction in the first fluid flow conduit such that the fluid flow can ingress to the axial earpiece bore of the earpiece from the external ear canal, thereby achieving an external ear canal pressure lesser than the ambient pressure.

(9) Another broad object of particular embodiments of the invention can be to provide an external ear canal pressure regulation device having a valved conduit which removably couples to the fluid flow generator and the earpiece. The valved conduit can be coupled in the first configuration with

the fluid flow generator and the earpiece to unidirectionally regulate the fluid flow in the first direction in the first fluid flow conduit. Additionally, the valved conduit can be coupled in the second configuration with the fluid flow generator and the earpiece to unidirectionally regulate the fluid flow in the second direction in the first fluid flow conduit.

(10) Another broad object of particular embodiments of the invention can be to provide a method of producing an external ear canal pressure regulation device, the method including providing a fluid flow generator capable of generating a fluid flow; providing a valved conduit capable of being fluidically coupled to the fluid flow generator, the valved conduit having a first fluid flow conduit; providing one or more valves capable of interrupting the first fluid flow conduit to unidirectionally regulate the fluid flow in the first fluid flow conduit; and providing an axial earpiece bore, which communicates between an earpiece first end and an earpiece second end of an earpiece, the axial earpiece bore capable of being fluidically coupled to the valved conduit opposite the fluid flow generator, the earpiece having a compliant earpiece external surface configured to sealably engage an external ear canal as a barrier between an external ear canal pressure and an ambient pressure.

(11) Another broad object of particular embodiments of the invention can be to provide a method of using an external ear canal pressure regulation device, the method including obtaining the external ear canal pressure regulation device including a fluid flow generator which generates a fluid flow; a valved conduit fluidically coupled to the fluid flow generator, the valved conduit having a first fluid flow conduit interruptible by one or more valves to unidirectionally regulate the fluid flow in the first fluid flow conduit; and an earpiece having an axial earpiece bore which communicates between an earpiece first end and an earpiece second end, the axial earpiece bore fluidically coupled to the valved conduit opposite the fluid flow generator, the earpiece having a compliant earpiece external surface configured to sealably engage an external ear canal as a barrier between an external ear canal pressure and an ambient pressure; sealably engaging the earpiece external surface of the earpiece with the external ear canal; generating the fluid flow between the fluid flow generator and the axial earpiece bore; and regulating an external ear canal pressure differential between the external ear canal pressure and the ambient pressure.

(12) Naturally, further objects of the invention are disclosed throughout other areas of the specification, drawings, and claims.

Description

III. BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is an illustration of a particular embodiment of an external ear canal pressure regulation device sealably engaged with the external ear canal.

(2) FIG. 1B is an illustration of a particular embodiment of an external ear canal pressure regulation device sealably engaged with the external ear canal.

(3) FIG. 2A is a schematic diagram of a particular embodiment of an external ear canal pressure regulation device operable to achieve an external ear canal pressure greater than the ambient pressure.

(4) FIG. 2B is a schematic diagram of a particular embodiment of an external ear canal pressure regulation device operable to achieve an external ear canal pressure lesser than the ambient pressure.

(5) FIG. 3A is a cross sectional view of a particular embodiment of a valve which can be utilized in embodiments of the external ear canal pressure regulation device.

(6) FIG. 3B is a cross sectional view of a particular embodiment of a valve which can be utilized in embodiments of the external ear canal pressure regulation device.

(7) FIG. 3C is a cross sectional view of a particular embodiment of a valve which can be utilized in embodiments of the external ear canal pressure regulation device.

(8) FIG. 3D is a cross sectional view of a particular embodiment of a valve which can be utilized in embodiments of the external ear canal pressure regulation device.

(9) FIG. 3E is a cross sectional view of a particular embodiment of a valve which can be utilized in embodiments of the external ear canal pressure regulation device.

(10) FIG. 4A is an illustration of a particular embodiment of an external ear canal pressure regulation device.

(11) FIG. 4B is an illustration of a particular embodiment of an external ear canal pressure regulation device.

(12) FIG. 4C is an illustration of a particular embodiment of an external ear canal pressure regulation device.

(13) FIG. 4D is an illustration of a particular embodiment of an external ear canal pressure regulation device.

(14) FIG. 4E is an illustration of a particular embodiment of an external ear canal pressure regulation device.

(15) FIG. 4F is an illustration of a particular embodiment of an external ear canal pressure regulation device.

(16) FIG. 4G is an illustration of the particular embodiment of the external ear canal pressure regulation device shown in FIG. 4F in which an earpiece uncouples from a fluid flow generator and a conduit body.

(17) FIG. 4H is an illustration of a particular embodiment of an external ear canal pressure regulation device.

(18) FIG. 4I is an illustration of a particular embodiment of an external ear canal pressure regulation device.

(19) FIG. 5 is a cross sectional view of a particular embodiment of an external ear canal pressure regulation device operable to achieve an external ear canal pressure greater than the ambient pressure.

(20) FIG. 6 is a cross sectional view of a particular embodiment of an external ear canal pressure regulation device operable to achieve an external ear canal pressure lesser than the ambient pressure.

(21) FIG. 7 is an exploded view of the particular embodiment of the external ear canal pressure regulation device shown in FIG. 5 operable to achieve an external ear canal pressure greater than the ambient pressure.

(22) FIG. 8 is an illustration of a first configuration of an external ear canal pressure regulation device operable to achieve an external ear canal pressure greater than the ambient pressure.

(23) FIG. 9 is an illustration of a method of reconfiguring the first configuration of the external ear canal pressure regulation device shown in FIG. 8 by end-to-end rotation of a conduit body.

(24) FIG. 10 is an illustration of a second configuration of the external ear canal pressure regulation device achieved by end-to-end rotation of a conduit body operable to achieve an external ear canal pressure lesser than the ambient pressure.

(25) FIG. 11A is an illustration of a first configuration of the external ear canal pressure regulation device having the earpiece external surface sealably engaged with the external ear canal.

(26) FIG. 11B is an illustration of the first configuration of the external ear canal pressure regulation device having the earpiece external surface sealably engaged with the external ear canal and having the fluid flow generator in a deformed condition generating sufficient pressure within the valved conduit to place a first valve in the open condition to provide a fluid flow toward the external ear canal to achieve an external ear canal pressure greater than the ambient pressure.

(27) FIG. 11C is an illustration of the first configuration of the external ear canal pressure regulation device having the earpiece external surface sealably engaged with the external ear canal and having the fluid flow generator returning toward a non-deformed condition which generates sufficient pressure to place the first valve in the closed condition to maintain the external ear canal

pressure and place a second valve in an open condition to generate a fluid flow from the ambient pressure toward the fluid flow generator.

(28) FIG. 11D is an illustration of the first configuration of the external ear canal pressure regulation device having the earpiece external surface sealably engaged with the external ear canal and having a third valve manually placed in the open condition by operation of a pressure relief element to generate a fluid flow from the external ear canal toward the ambient pressure to return the external ear canal pressure toward the ambient pressure.

(29) FIG. 12A is an illustration of a second configuration of the external ear canal pressure regulation device having the earpiece external surface sealably engaged with the external ear canal.

(30) FIG. 12B is an illustration of the second configuration of the external ear canal pressure regulation device having the earpiece external surface sealably engaged with the external ear canal and having the fluid flow generator in a deformed condition generating sufficient pressure within the valved conduit to place the second valve in the open condition to provide a fluid flow toward the ambient pressure while maintaining the external ear canal pressure at the ambient pressure.

(31) FIG. 12C is an illustration of the second configuration of the external ear canal pressure regulation device having the earpiece external surface sealably engaged with the external ear canal and having the fluid flow generator returning toward a non-deformed condition which generates sufficient pressure to place the first valve in the open condition and the second valve in the closed condition to generate a fluid flow from the external ear canal toward the fluid flow generator to generate an external ear canal pressure less than the ambient pressure.

(32) FIG. 12D is an illustration of the second configuration of the external ear canal pressure regulation device having the earpiece external surface sealably engaged with the external ear canal and having a third valve manually placed in the open condition by operation of a pressure relief element to generate a fluid flow from the ambient pressure toward the external ear canal to return the external ear canal pressure toward the ambient pressure.

(33) FIG. 13A is a top view of a particular embodiment of an external ear canal pressure regulation device.

(34) FIG. 13B is a side view of a particular embodiment of an external ear canal pressure regulation device.

(35) FIG. 13C is an exploded view of a particular embodiment of an external ear canal pressure regulation device.

(36) FIG. 13D is an exploded view of a particular embodiment of an external ear canal pressure regulation device.

(37) FIG. 13E is a cross sectional view of the particular embodiment of the external ear canal pressure regulation device shown in FIG. 13A operable to generate an external ear canal pressure greater than the ambient pressure.

(38) FIG. 13F is a cross sectional view of a particular embodiment of the external ear canal pressure regulation device operable to generate an external ear canal pressure lesser than the ambient pressure.

(39) FIG. 14A is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the first configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator generates and maintains a substantially invariant external ear canal pressure greater than the ambient pressure at a maximum pressure for a time period and by operation of the pressure relief element, the external ear canal pressure returns toward the ambient pressure.

(40) FIG. 14B is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the first configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure greater than the ambient pressure with the pressure wave being a sine wave having smooth repetitive periodic

oscillations.

(41) FIG. 14C is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the first configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure greater than the ambient pressure with the pressure wave being a truncated wave in which the apex of the pressure wave has a constant pressure over a time period.

(42) FIG. 14D is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the first configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure greater than the ambient pressure with the pressure wave being a triangle wave having linear leading and trailing edges.

(43) FIG. 14E is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the first configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure greater than the ambient pressure with the pressure wave being a sawtooth wave in which the leading edge changes pressure over a time period which is greater than the time period in which the trailing edge changes pressure.

(44) FIG. 14F is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the first configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure greater than the ambient pressure with the pressure wave being a truncated wave in which the apex of the pressure wave has a constant pressure over a time period.

(45) FIG. 14G is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the first configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure greater than the ambient pressure.

(46) FIG. 15A is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the second configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator generates and maintains a substantially invariant external ear canal pressure lesser than the ambient pressure at a maximum pressure for a time period and by operation of the pressure relief element, the external ear canal pressure returns toward the ambient pressure.

(47) FIG. 15B is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the second configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure lesser than the ambient pressure with the pressure wave being a sine wave having smooth repetitive periodic oscillations.

(48) FIG. 15C is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the second configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure lesser than the ambient pressure with the pressure wave being a truncated wave in which the apex of the pressure wave has a constant pressure over a time period.

(49) FIG. 15D is a plot of the external ear canal pressure relative to the ambient pressure achieved

over a time period which represents a method of use of the second configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure lesser than the ambient pressure with the pressure wave being a triangle wave having linear leading and trailing edges.

(50) FIG. 15E is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the second configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure lesser than the ambient pressure with the pressure wave being a reverse sawtooth wave in which the leading edge changes pressure over a time period which is lesser than the time period in which the trailing edge changes pressure.

(51) FIG. 15F is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the second configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure lesser than the ambient pressure with the pressure wave being a truncated wave in which the apex of the pressure wave has a constant pressure over a time period.

(52) FIG. 15G is a plot of the external ear canal pressure relative to the ambient pressure achieved over a time period which represents a method of use of the second configuration of the external ear canal pressure regulation device in which operation of the fluid flow generator with intermittent operation of the pressure relief element generates a pulsatile external ear canal pressure lesser than the ambient pressure.

IV. MODE(S) FOR CARRYING OUT THE INVENTION

(53) Now referring primarily to FIG. 1A and FIG. 1B, which illustrate particular methods of using an external ear canal pressure regulation device (1) including one or more of: a fluid flow generator (2), an earpiece (3) having an axial earpiece bore (4), and a valved conduit (5) fluidically coupled to the fluid flow generator (2) and the axial earpiece bore (4). The method of use can include sealably engaging an external ear canal (6) with an earpiece external surface (7) of the earpiece (3), generating a fluid flow (8) between the fluid flow generator (2) and the axial earpiece bore (4), and regulating a pressure differential (9) between an external ear canal pressure (10) and an ambient pressure (11) to alleviate one or more disorder symptoms or treat one or more disorders.

(54) The term “pressure differential” for the purposes of this invention means the difference in pressure between two locations.

(55) The term “pressure differential amplitude” for the purposes of this invention means the numerical value of the difference in pressure between two locations. The pressure differential amplitude (59) can be expressed as a number without a sign (positive or negative), regardless of whether the pressure is greater or lesser in the first location relative to the second location. As an illustrative example, an external ear canal pressure (10) of +50 kilopascals above the ambient pressure (11) and an external ear canal pressure (10) of -50 kilopascals below the ambient pressure (11) can both have a pressure differential amplitude (59) of 50 kilopascals.

(56) The term “external ear canal pressure” for the purposes of this invention means forces exerted within the external ear canal (6) and, without limitation to the breadth of the foregoing, means forces exerted within the external ear canal (6) by a fluid volume (12), a pre-selected fluid volume (12), or a fluid flow (8) delivered to or generated in the external ear canal (6) by operation of the external ear canal pressure regulation device (1).

(57) The term “ambient pressure” for the purposes of this invention means forces exerted external to the external ear canal (6) in the ambient environment and, without limitation to the breadth of the foregoing, means forces exerted on the earpiece (3) having the earpiece external surface (7) sealably engaged with the external ear canal (6), as herein described.

(58) The term “pre-selected” for the purposes of this invention means a parameter, such as a fluid volume (12) or a pressure differential amplitude (59) which has been determined prior to administration, for example by a user (23) of the external ear canal pressure regulation device (1), for delivery to, generation in, or administration to the external ear canal (6) by operation of the external ear canal pressure regulation device (1) and subsequently delivered to, generated in, or administered to the external ear canal (6) by operation of the external ear canal pressure regulation device (1). For example, a pre-selected fluid volume (12) of 10 milliliters can be prior selected for delivery to the external ear canal (6) by operation of the external ear canal pressure regulation device (1) and subsequently, the pre-selected fluid volume (12) of 10 milliliters can be delivered to the external ear canal (6) by operation of the external ear canal pressure regulation device (1).

(59) The term “symptom” for the purposes of this invention means any discomfort or combination of discomforts associated with a disorder. Without limiting the breadth of the foregoing, symptoms can include: dizziness; vertigo; nausea; imbalance; paresthesia; dysesthesia; sensitivity to light; sensitivity to odor; sensitivity to sound; anxiety; sleeplessness; irritability; fatigue; loss of appetite; blurred vision; gut disturbances; acute pain or chronic pain of varying characteristics including but not limited to throbbing, tearing, sharp, dull, deep, lancinating, burning, aching, stabbing, intense, lightning-like, sense of swelling, or tingling; or the like; or combinations thereof.

(60) The term “disorder” for the purposes of this invention means a physical or mental condition which may not be normal or healthy. Without limiting the breadth of the foregoing, a disorder can include: neuropathic craniofacial pain syndromes such as neuralgias, for example trigeminal neuralgia; temporomandibular joint syndrome; headache syndromes such as migraine headaches, chronic daily headaches, cluster headaches, muscle tension headaches, post-traumatic headaches, or chronic paroxysmal hemicranias; endolymphatic hydrops; vertigo; tinnitus; syndromes resulting from brain injury; syndromes resulting from impaired neurologic function, including cognitive disorders such as attention deficit disorder, emotional disorders such as anxiety disorders, or seizure disorders; phantom limb; middle ear disorders; inner ear disorders; or the like, or combinations thereof.

(61) Now referring primarily to FIG. 2A and FIG. 2B, the fluid flow generator (2) can have any of a wide variety of numerous configurations capable of generating a fluid flow (8) between the fluid flow generator (2) and the axial earpiece bore (4). As to particular embodiments, the fluid flow generator (2) can include a volumetrically adjustable element (13) capable of adjusting between a greater volume and a lesser volume. As an illustrative example, adjusting the volumetrically adjustable element (13) from a greater volume to a lesser volume can generate a fluid flow (8) away from the fluid flow generator (2) whereas adjusting the volumetrically adjustable element (13) from a lesser volume to a greater volume can generate a fluid flow (8) toward the fluid flow generator (2).

(62) As to particular embodiments, the fluid flow generator (2) can include a bladder (14) or a diaphragm (15) which has a resiliently flexible wall (16) having a wall external surface (17) and a wall internal surface (18). The wall external surface (17) can be configured in any manner which allows deformation of the resiliently flexible wall (16) (as shown in the examples of FIG. 11A through FIG. 12D and FIG. 13A through FIG. 13F). The wall internal surface (18) can define an internal volume (19) (whether in whole or in part as an assembly with the valved conduit (5)). The resiliently flexible wall (16) in a deformed condition (20) (as shown in the examples of FIG. 11B and FIG. 12B) can decrease the internal volume (19), and in return toward a non-deformed condition (21) (as shown in the example of FIG. 11C and FIG. 12C), can increase the internal volume (19). The change in the internal volume (19) can generate a fluid flow (8) between the fluid flow generator (2) and the axial earpiece bore (4), which can be regulated by the valved conduit (5). As to particular embodiments, the bladder (14) or the diaphragm (15) can have an internal volume (19) in the non-deformed condition (21) which can be insufficient upon complete reduction in internal volume (19) to the deformed condition (20) to generate a fluid flow (8) or an amount of

pressure (22) capable of causing discomfort to a user (23) of the external ear canal pressure regulation device (1) or injury to an auditory meatus (24) or a tympanic membrane (25).

(63) As to other particular embodiments, the fluid flow generator (2) can include a positive displacement pump (26) in which a piston (27) reciprocally operates in a barrel (28) (as shown in the illustrative example of FIG. 4C) to adjust a barrel internal volume between a lesser volume and a greater volume. The reciprocal motion of the piston (27) within the barrel (28) can generate a fluid flow (8) between the fluid flow generator (2) and the axial earpiece bore (4), which can be regulated by the valved conduit (5). As to particular embodiments, the barrel (28) can have a barrel internal volume which can be insufficient upon complete reduction in barrel internal volume by travel of the piston (27) within the barrel (28) to generate a fluid flow (8) or an amount of pressure (22) capable of causing discomfort to the user (23) of the external ear canal pressure regulation device (1) or injury to the auditory meatus (24) or the tympanic membrane (25).

(64) The fluid flow generator (2) can be configured to generate a fluid flow (8) in the valved conduit (5) between the fluid flow generator (2) and the axial earpiece bore (4), whereby the fluid flow (8) can have a fluid volume (12) typically in a range of between 0 milliliters to about 20 milliliters; however, embodiments can have a lesser or greater fluid volume (12) depending upon the application. As to particular embodiments, the fluid volume (12) can be a pre-selected fluid volume (12), which can be selected from one or more of the group including or consisting of: between 0 milliliters to about 2 milliliters, between about 1 milliliter to about 3 milliliters, between about 2 milliliters to about 4 milliliters, between about 3 milliliters to about 5 milliliters, between about 4 milliliters to about 6 milliliters, between about 5 milliliters to about 7 milliliters, between about 6 milliliters to about 8 milliliters, between about 7 milliliters to about 9 milliliters, between about 8 milliliters to about 10 milliliters, between about 9 milliliters to about 11 milliliters, between about 10 milliliters to about 12 milliliters, between about 11 milliliters to about 13 milliliters, between about 12 milliliters to about 14 milliliters, between about 13 milliliters to about 15 milliliters, between about 14 milliliters to about 16 milliliters, between about 15 milliliters to about 17 milliliters, between about 16 milliliters to about 18 milliliters, between about 17 milliliters to about 19 milliliters, and between about 18 milliliters to about 20 milliliters.

(65) One or more pre-selected fluid volumes (12) can be generated with the external ear canal pressure regulation device (1) depending upon the method of use, which can be further influenced by factors such as user (23) anatomy, physiology, or biochemistry of the auditory meatus (24); disorder symptom targeted for alleviation; disorder targeted for treatment; observable effect(s) of using one or more pre-selected fluid volumes (12) in a particular method of using the external ear canal pressure regulation device (1); or the like; or combinations thereof; but not so much as to cause discomfort to the user (23) or injury to the auditory meatus (24) or the tympanic membrane (25).

(66) Again referring primarily to FIGS. 2A and 2B, the earpiece (3) can have a compliant earpiece external surface (7) configured to insert into the external ear canal (6) of the auditory meatus (24), thus acting as a barrier between the external ear canal pressure (10) and the ambient pressure (11). Embodiments of the earpiece (3) can be configured to sufficiently sealably engage with the external ear canal (6) to resist axial or lateral displacement in view of normal anatomical variations of the external ear canal (6) over a normal range of operating temperatures of between about 20° C. (about 68° F.) to about 50° C. (about 122° F.) and allow generation and maintenance of a normal range of operating pressures of between about negative 50 kilopascals (−50 kPa) below the ambient pressure (11) to about positive 50 kilopascals (+50 kPa) above the ambient pressure (11).

(67) The earpiece (3) of the external ear canal pressure regulation device (1) can be formed from a compliant material which can compressibly deform upon engagement with the external ear canal (6), thereby allowing the earpiece (3) to sealably conform to the external ear canal (6). As to these particular embodiments, the earpiece (3) can be formed, molded, three-dimensionally printed, or otherwise fabricated from any of a numerous and wide variety of materials capable of sealable

engagement with the external ear canal (6), including or consisting of: a silicone, a foam (including polyurethane foam), a polyvinylsiloxane, a low durometer elastomer, or the like, or combinations thereof.

(68) As to particular embodiments, the earpiece (3) can be formed from one material, for example a lesser durometer elastomer. As to other particular embodiments, the earpiece (3) can be formed from a plurality of layers, for example an inner core layer having a greater durometer surrounded by an outer layer having a lesser durometer or an inner core layer having a lesser durometer surrounded by an outer layer having a greater durometer. As to yet other particular embodiments, a flexible earpiece wall can define a hollow inner space of the earpiece (3), whereby the flexible earpiece wall can deform to allow the earpiece external surface (7) to sealably conform to the external ear canal (6).

(69) As to particular embodiments, a portion of the earpiece external surface (7) can inwardly taper approaching an earpiece second end (29) (as shown in the examples of FIG. 4A through FIG. 4D).

As an illustrative example of particular embodiments of this configuration, the earpiece external surface (7) can be configured in the general form of a truncated cone inwardly tapering approaching the earpiece second end (29) (as shown in the examples of FIG. 4E through FIG. 4G).

(70) The earpiece external surface (7) can further include a plurality of circumferential ribs (30) disposed in spaced apart relation between an earpiece first end (31) and the earpiece second end (29). Each of the plurality of circumferential ribs (30) can extend from the earpiece external surface (7) a substantially uniform height; however, as to those embodiments of the earpiece external surface (7) having a conical configuration, the plurality of circumferential ribs (30) can have a rib diameter (32) which decreases approaching the earpiece second end (29) (as shown in example of FIG. 4H). As an illustrative example, the rib diameter (32) of a first circumferential rib (33) proximate the earpiece first end (31) can be about seven centimeters and the rib diameter (32) of a last circumferential rib (34) proximate the earpiece second end (29) can be about four centimeters, with the circumferential ribs (30) disposed between the first circumferential rib (33) and the last circumferential rib (34) having rib diameters (32) which respectively decrease from the first circumferential rib (33) to the last circumferential rib (34). However, embodiments need not necessarily be so limited and the plurality of circumferential ribs (30) can be configured in any of a wide variety of numerous configurations adapted to insert into and sealably engage with the external ear canal (6), thus acting as a barrier between the external ear canal pressure (10) and the ambient pressure (11).

(71) The earpiece external surface (7) can remain sealably engaged with the external ear canal (6) by frictional forces between the earpiece external surface (7) and the external ear canal (6). As to particular embodiments, the earpiece external surface (7) can remain engaged with the external ear canal (6) by forcible urging against the external ear canal pressure regulation device (1) during normal operation. As to other particular embodiments, a restraint element coupled to the external ear canal pressure regulation device (1) can be worn about the ear (35) or the head (36) to assist with retention of the earpiece (3) within the external ear canal (6).

(72) Again referring primarily to FIG. 2A and FIG. 2B, the earpiece (3) can have an axial earpiece bore (4) which communicates between the earpiece first end (31) and the earpiece second end (29). The axial earpiece bore (4) proximate the earpiece first end (31) can be fluidically coupled to a valved conduit (5) and configured to allow a fluid flow (8) between the earpiece first and second ends (31)(29).

(73) Now referring primarily to FIG. 5 and FIG. 6, the earpiece (3) can further include a tubular bolt (37) disposed about the axial earpiece bore (4). The tubular bolt (37) can communicate, whether in whole or in part, between the earpiece first end (31) and the earpiece second end (29), providing a bolt bore (38) communicating, whether in whole or in part, between the earpiece first end (31) and the earpiece second end (29). The tubular bolt (37) can be sufficiently rigid to reduce or prevent deformation of the axial earpiece bore (4) upon sealable engagement of the earpiece

external surface (7) with the external ear canal (6) to maintain sufficient fluid flow (8) within the axial earpiece bore (4) during normal use as described above, for example over a normal range of operating temperatures and a normal range of operating pressures.

(74) The tubular bolt (37) can further include a bolt external surface (39) dimensioned for removable insertion into the axial earpiece bore (4), providing an adequate fluid-tight seal to maintain sufficient fluid flow (8) within the axial earpiece bore (4) during normal use as described above, for example over a normal range of operating temperatures and a normal range of operating pressures. As to particular embodiments, the bolt external surface (39) can further include a plurality of circumferential barbs spaced apart along the bolt external surface (39) to assist in retaining the tubular bolt (37) within the axial earpiece bore (4) and in providing the fluid-tight seal.

(75) As to particular embodiments including a discrete tubular bolt (37), the earpiece (3) and the tubular bolt (37) can be provided as a one-piece construct having the earpiece (3) molded or formed about the tubular bolt (37). As to other particular embodiments, the earpiece (3) can be formed or molded to provide increasing rigidity approaching the axial earpiece bore (4). However, embodiments need not necessarily be so limited, as any of a wide variety of numerous structures known to those of ordinary skill in the art can be utilized to provide a tubular bolt (37) in fluid-tight relation with the axial earpiece bore (4).

(76) Now referring primarily to FIG. 7 through FIG. 10, the external ear canal pressure regulation device (1) can include a valved conduit (5) fluidically coupled to the fluid flow generator (2) and the axial earpiece bore (4). As to particular embodiments, the valved conduit (5) can be included in a conduit body (41) having a configuration which removably couples to the fluid flow generator (2) and the earpiece (3), whereby the fluid flow generator (2) and the earpiece (3) can be removably coupled to either one of a conduit body first end (42) or a conduit body second end (43). The releasably couplable surfaces (44) of the fluid flow generator (2), the earpiece (3), the conduit body first end (42), and the conduit body second end (43) can have sufficiently similar configurations (as shown in the example of FIG. 7) to allow the fluid flow generator (2) and the earpiece (3) to releasably couple to either of the conduit body first end (42) or the conduit body second end (43) depending upon whether the valved conduit (5) operates to achieve a pressure differential (9) having the external ear canal pressure (10) greater than the ambient pressure (11) (as shown in the example of FIG. 5) or whether the valved conduit (5) operates to achieve a pressure differential (9) having the external ear canal pressure (10) lesser than ambient the pressure (11) (as shown in the example of FIG. 6).

(77) The releasably couplable surfaces (44) of the fluid flow generator (2), the earpiece (3), the conduit body first end (42), and the conduit body second end (43) can matably engage. As an illustrative example, the releasably couplable surfaces (44) can be configured as rotatably matable spiral threads. However, embodiments need not necessarily be so limited and can have releasably couplable surfaces (44) configured in any of a wide variety of numerous manners which allow the conduit body (41) including the valved conduit (5) to be positioned in a first configuration (45) to operationally achieve a pressure differential (9) having the external ear canal pressure (10) greater than the ambient pressure (11) (as shown in the example of FIG. 5) and further allows the conduit body (41) including the valved conduit (5) to be positioned in a second configuration (46) to operationally achieve a pressure differential (9) having the external ear canal pressure (10) lesser than the ambient pressure (11) (as shown in the example of FIG. 6).

(78) As an illustrative example, the conduit body (41) including the valved conduit (5) can be positioned in the first configuration (45) by removably coupling the conduit body first end (42) with the fluid flow generator (2) and removably coupling the conduit body second end (43) with the earpiece (3) (as shown in the example of FIG. 8). As such, the fluid flow (8) can be regulated in a first direction (47) in the valved conduit (8) to operationally achieve a pressure differential (9) having the external ear canal pressure (10) greater than the ambient pressure (11) (as shown in the

example of FIG. 5).

(79) As to particular embodiments, the fluid flow generator (2) can be removed from the conduit body first end (42) and the earpiece (3) can be removed from the conduit body second end (43) to disassemble the first configuration (45). The conduit body (41) can be rotated without any structural alteration to reverse orientation of the conduit body first and second ends (42)(43) (as shown in the example of FIG. 9). The valved conduit (5) can be positioned in the second configuration (46) by removably coupling the conduit body first end (42) with the earpiece (3) and removably coupling the conduit body second end (43) with the fluid flow generator (2) (as shown in the example of FIG. 10). As such, the fluid flow (8) can be regulated in a second direction (48) in the valved conduit (5) to operationally achieve a pressure differential (9) having the external ear canal pressure (10) lesser than the ambient pressure (11) (as shown in the example of FIG. 6).

(80) Now referring primarily to FIG. 2A, FIG. 2B, FIG. 5, and FIG. 6, the valved conduit (5) can include a first fluid flow conduit (49) communicating between a first fluid flow conduit first end (50) and a first fluid flow conduit second end (51). As to the particular embodiments of the external ear canal pressure regulation device (1) described herein, the first fluid flow conduit first end (50) communicates with the fluid flow generator (2) and the first fluid flow conduit second end (51) communicates with the axial earpiece bore (4) of the earpiece (3), whether the conduit body (41) is positioned in the first configuration (45) or the second configuration (46).

(81) The first fluid flow conduit (49) can be interruptible by a first valve (52) to unidirectionally regulate the fluid flow (8) between the first fluid flow conduit first and second ends (50)(51) and, correspondingly, between the fluid flow generator (2) and the axial earpiece bore (4). In the first configuration (45) described above, the fluid flow generator (2) can be sealably engaged with the first fluid flow conduit first end (50) and the axial earpiece bore (4) can be sealably engaged with the first fluid flow conduit second end (51) to unidirectionally regulate the fluid flow (8) in the first direction (47) from the fluid flow generator (2) toward the axial earpiece bore (4). As such, the external ear canal pressure regulation device (1) sealably engaged with an external ear canal (6) can operationally achieve a pressure differential (9) having the external ear canal pressure (10) greater than the ambient pressure (11) by transferring a fluid volume (12) from the fluid flow generator (2) toward the external ear canal (6). In the second configuration (46) described above, the fluid flow generator (2) can be sealably engaged with the first fluid flow conduit first end (50) and the axial earpiece bore (4) can be sealably engaged with the first fluid flow conduit second end (51) to unidirectionally regulate the fluid flow (8) in the second direction (48) from the axial earpiece bore (4) toward the fluid flow generator (2). As such, the external ear canal pressure regulation device (1) sealably engaged with an external ear canal (6) can operationally achieve a pressure differential (9) having the external ear canal pressure (10) lesser than the ambient pressure (11) by transferring a fluid volume (12) from the external ear canal (6) toward the fluid flow generator (2).

(82) Again referring primarily to FIG. 2A, FIG. 2B, FIG. 5, and FIG. 6, the first valve (52) can divide the first fluid flow conduit (49) into a first portion (53) proximate the first fluid flow conduit first end (50) and, correspondingly, proximate the fluid flow generator (2), and a second portion (54) proximate the first fluid flow conduit second end (51), and, correspondingly, proximate the axial earpiece bore (4) of the earpiece (3). As to particular embodiments, the valved conduit (5) can further include a second fluid flow conduit (55) fluidically coupled between the first portion (53) of the first fluid flow conduit (49) and the ambient pressure (11). As to the particular embodiments of the external ear canal pressure regulation device (1) described herein, the second fluid flow conduit (55) fluidically couples to the first portion (53) of the first fluid flow conduit (49) proximate the fluid flow generator (2), whether the conduit body (41) is positioned in the first configuration (45) or the second configuration (46). The second fluid flow conduit (55) can be interruptible by a second valve (56) to unidirectionally regulate the fluid flow (8) in the second fluid flow conduit (55).

(83) Now referring primarily to FIG. 2A and FIG. 5, as to particular embodiments of the first configuration (45) having a fluid flow generator (2) which includes a volumetrically adjustable

element (13) having an internal volume (19) bounded by a resiliently flexible wall (16), a deformed condition (20) of the resiliently flexible wall (16) can decrease the internal volume (19) to generate a fluid flow (8) in the first direction (47) in the first fluid flow conduit (49) from the fluid flow generator (2) toward the axial earpiece bore (4) of the earpiece (3), whereby the first valve (52) and the second valve (56) unidirectionally regulate the fluid flow (8) to egress from the axial earpiece bore (4). As such, the external ear canal pressure regulation device (1) sealably engaged with an external ear canal (6) can operationally achieve a pressure differential (9) having the external ear canal pressure (10) greater than the ambient pressure (11) by transferring a fluid volume (12) from the fluid flow generator (2) toward the external ear canal (6).

(84) The resiliently flexible wall (16) of the volumetrically adjustable element (13) can return to a non-deformed condition (21) which can increase the internal volume (19) to generate a fluid flow (8) in the second fluid flow conduit (55), whereby the second valve (56) unidirectionally regulates the fluid flow (8) to ingress from an ambient pressure (11) toward the fluid flow generator (2). The first valve (52) can interrupt the fluid flow (8) in the first fluid flow conduit (49) from the axial earpiece bore (4) toward the fluid flow generator (2). Embodiments of the external ear canal pressure regulation device (1) sealably engaged with an external ear canal (6) can operationally maintain a pressure differential (9) in which the external ear canal pressure (10) can be maintained greater than the ambient pressure (11) and concurrently transfer a fluid volume (12) from the ambient pressure (11) toward the fluid flow generator (2) to return the resiliently flexible wall (16) of the volumetrically adjustable element (13) toward the non-deformed condition (21).

(85) Now referring primarily to FIG. 2B and FIG. 6, as to particular embodiments of the second configuration (46) having a fluid flow generator (2) which includes a volumetrically adjustable element (13) having an internal volume (19) bounded by a resiliently flexible wall (16), a deformed condition (20) of the resiliently flexible wall (16) can decrease the internal volume (19) to generate a fluid flow (8) in the second fluid flow conduit (55) from the fluid flow generator (2) toward the ambient pressure (11), whereby the first valve (52) and the second valve (56) unidirectionally regulate the fluid flow (8) to egress from the second fluid flow conduit (55) toward the ambient pressure (11).

(86) The resiliently flexible wall (16) of the volumetrically adjustable element (13) can return to a non-deformed condition (21) which can increase the internal volume (19) to generate a fluid flow (8) in the second direction (48) in the first fluid flow conduit (49), whereby the first valve (52) unidirectionally regulates the fluid flow (8) to ingress from the axial earpiece bore (4) of the earpiece (3) toward the fluid flow generator (2). The second valve (56) can interrupt the fluid flow (8) in the second fluid flow conduit (55) from the ambient pressure (11) toward the fluid flow generator (2). As such, the external ear canal pressure regulation device (1) sealably engaged with an external ear canal (6) can operationally achieve and maintain a pressure differential (9) in which the external ear canal pressure (10) can be maintained lesser than the ambient pressure (11) and concurrently transfer a fluid volume (12) from the external ear canal (6) toward the fluid flow generator (2) to return the resiliently flexible wall (16) of the volumetrically adjustable element (13) toward the non-deformed condition (21).

(87) Now referring primarily to FIG. 2A, FIG. 2B, FIG. 5, and FIG. 6, as to particular embodiments, the valved conduit (5) can further include a third fluid flow conduit (57) fluidically coupled between the second portion (54) of the first fluid flow conduit (49) and the ambient pressure (11). As to the particular embodiments of the external ear canal pressure regulation device (1) described herein, the third fluid flow conduit (57) fluidically couples to the second portion (54) of the first fluid flow conduit (49) proximate the axial earpiece bore (4), whether the conduit body (41) is positioned in the first configuration (45) or the second configuration (46).

(88) The third fluid flow conduit (57) can be interruptible by a third valve (58) to unidirectionally regulate the fluid flow (8) in the third fluid flow conduit (57). As to particular embodiments, the third valve (58) can regulate the fluid flow (8) to egress toward the ambient pressure (11) from the

third fluid flow conduit (57). As to other particular embodiments, the third valve (58) can regulate the fluid flow (8) to ingress from the ambient pressure (11) toward the third fluid flow conduit (57). (89) As to particular embodiments, the third valve (58) can interrupt the fluid flow (8) within the third fluid flow conduit (57) until a pressure differential (9) between the second portion (54) of the first fluid flow conduit (49) and the ambient pressure (11) exceeds a pre-selected pressure differential (9) having a pressure differential amplitude (59) typically in a range of between 0 kilopascals to about 50 kilopascals; however embodiments can have a lesser or greater pre-selected pressure differential amplitude (59) depending upon the application. As to particular embodiments, the pre-selected pressure differential amplitude (59) can be selected from the group including of consisting of: between 0 kilopascals to about 5 kilopascals, between about 2.5 kilopascals to about 7.5 kilopascals, between about 5 kilopascals to about 10 kilopascals, between about 7.5 kilopascals to about 12.5 kilopascals, between about 10 kilopascals to about 15 kilopascals, between about 12.5 kilopascals to about 17.5 kilopascals, between about 15 kilopascals to about 20 kilopascals, between about 17.5 kilopascals to about 22.5 kilopascals, between about 20 kilopascals to about 25 kilopascals, between about 22.5 kilopascals to about 27.5 kilopascals, between about 25 kilopascals to about 30 kilopascals, between about 27.5 kilopascals to about 32.5 kilopascals, between about 30 kilopascals to about 35 kilopascals, between about 32.5 kilopascals to about 37.5 kilopascals, between about 35 kilopascals to about 40 kilopascals, between about 37.5 kilopascals to about 42.5 kilopascals, between about 40 kilopascals to about 45 kilopascals, between about 42.5 kilopascals to about 47.5 kilopascals, and between about 45 kilopascals to about 50 kilopascals.

(90) One or more pre-selected pressure differential amplitudes (59) can be generated with the external ear canal pressure regulation device (1) depending upon the method of use, which can be further influenced by factors such as user (23) anatomy, physiology, or biochemistry of the auditory meatus (24); disorder symptom targeted for alleviation; disorder targeted for treatment; observable effect(s) of using one or more pre-selected pressure differential amplitudes (59) in a particular method of using the external ear canal pressure regulation device (1); or the like; or combinations thereof; but not so much as to cause discomfort to the user (23) or injury to the auditory meatus (24) or the tympanic membrane (25).

(91) Now referring primarily to FIG. 2A and FIG. 2B, the valved conduit (5) can include a fluid flow manifold (60) interruptible by operation of one or more valves (61), for example the first valve (52), the second valve (56), the third valve (58), or additional valves (61) to correspondingly alter the configuration of a manifold fluid flow path (62) within the fluid flow manifold (60) to regulate the fluid flow (8) within the fluid flow manifold (60). While the figures schematically illustrate particular configurations of the fluid flow manifold (60) which correspondingly define particular configurations of the manifold fluid flow path (62), these embodiments need not necessarily be so limited in regard to the configuration of the fluid flow manifold (60) or the manifold fluid flow path (62) of the valved conduit (5) and embodiments can include any of a wide variety of numerous configurations which can fluidically couple the first, second, or third fluid flow conduits (49)(55)(57) as above described (or additional fluid flow conduits) whether as a plurality of discrete conduits, a one-piece manifold, or defined by a conduit body (41) whether formed, molded, three-dimensionally printed, or otherwise fabricated as a one-piece construct or assembled from a plurality of pieces into which the first, second, or third valves (52)(56)(58) (or additional valves (61)) can be disposed, assembled, or otherwise coupled to generate the valved conduit (5) and without limitation to the breadth of the foregoing, include the configurations of the conduit body (41) shown in illustrative examples of FIG. 4A through FIG. 6, which can be injection molded as a one-piece conduit body (41) about or which can receive the first, second, or third valves (52)(56)(58) (or additional valves).

(92) Now referring primarily to FIG. 3A through FIG. 3E, the first, second, or third valves (52)(56)(58) schematically illustrated in FIG. 2A and FIG. 2B can have any type of valve configuration capable of regulating a fluid flow (8) as described herein and without limitation to the breadth of

the foregoing, can include a spring-loaded ball check valve (63) (as shown in the example of FIG. 3C), a flapper valve (64) having a hinge (94) (as shown in the example of FIG. 3B), a spring-loaded valve having a seat and seatable deformable circular lip (65) (as shown in the example of FIG. 3A), an umbrella valve (66) (as shown in the example of FIG. 3D), a duckbill valve (67) (as shown in the example of FIG. 3E), or other valves (61) which can operate between a closed condition (68) and an open condition (69) to unidirectionally regulate fluid flow (8) in a pre-selected range.

(93) As to particular embodiments, each of the first, second, or third valves (52)(56)(58) can operate between the closed condition (68), which can be substantially leak-tight to backward flow and substantially leak-tight to forward fluid flow (8) up to about a 50 kilopascal pressure differential amplitude (59) on opposed sides of the valve (61), and the open condition (69), which can have a forward flow in the range of about 0.2 milliliters per second to about 20 milliliters per second. As to particular embodiments, the pressure differential (9) between opposed sides of a valve (61) or the forward fluid flow (8) in the open condition (69) of a valve (61) can be adjusted by the configuration of the valve (61), the unrestricted cross-sectional area of the manifold fluid flow path (62), or the like, or combinations thereof. Additionally, while examples of the external ear canal pressure regulation device (1) disclosed can generate a pressure differential amplitude (59) between the external ear canal pressure (10) and the ambient pressure (11) of up to 50 kilopascals, these examples are not intended to teach or suggest that all embodiments of the external ear canal pressure regulation device (1) necessarily achieve this pressure differential amplitude (59) between the external ear canal pressure (10) and the ambient pressure (11). Rather, particular embodiments of the external ear canal pressure regulation device (1) can be configured to achieve a pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11) based on being effective to alleviate one or more disorder symptoms, for example neurologically-mediated pain, or treat one or more disorders, for example craniofacial pain syndromes or headache syndromes.

(94) Now referring primarily to FIG. 5, FIG. 6, and FIG. 11A through FIG. 12D, as to particular embodiments, a valve (61) can be operatively coupled to a pressure relief element (70) configured to allow manual operation of the valve (61) between the closed condition (68) and the open condition (69). As an illustrative example, a pressure relief element (70) can be operable to allow the external ear canal pressure (10) to return toward the ambient pressure (11), whether from an external ear canal pressure (10) greater than the ambient pressure (11) or an external ear canal pressure (10) lesser than the ambient pressure (11). As to particular embodiments, a pressure relief element (70) can be operable to allow a fluid volume (12) to egress from a portion of the valved conduit (5) toward the ambient pressure (11) or ingress from the ambient pressure (11) toward a portion of the valved conduit (5) upon reaching or exceeding a pre-selected threshold external ear canal pressure (10), thereby reducing the risk of discomfort to a user (23) or injury to the auditory meatus (24) or the tympanic membrane (25).

(95) As to particular embodiments, the pressure relief element (70) can be configured to extend a sufficient distance outward from the conduit body (41) to allow gripping engagement by a user (23). As to other particular embodiments, the pressure relief element (70) can be configured as a resiliently flexible portion of the conduit body (41) which can flex upon pressing engagement, placing the valve (61) in the open condition (69). Upon disengagement of the pressure relief element (70), the valve (61) can return to the closed condition (68).

(96) Again referring primarily to FIG. 5, FIG. 6, and FIG. 11A through FIG. 12D, as to particular embodiments, a second valve pressure relief element (71) can be coupled to the second valve (56) and a third valve pressure relief element (72) can be coupled to the third valve (58). Each of the second and third valve pressure relief elements (71)(72) can be manually operable to correspondingly generate a fluid flow (8) in the second or third fluid flow conduits (55)(57), respectively.

(97) Now referring primarily to FIG. 11A through FIG. 12D, as to particular embodiments, the external ear canal pressure regulation device (1) can be configured to achieve an external ear canal pressure (10) which can be lesser or greater than the ambient pressure (11). The effective amount of external ear canal pressure (10) to alleviate one or more disorder symptoms or treat one or more disorders, or the greatest amount of external ear canal pressure (10) achieved in a pressure regulation profile (73) generated by an embodiment of the external ear canal pressure regulation device (1) can have a range from just above or just below the ambient pressure (11) to just above or below the external ear canal pressure (10) at which discomfort may occur to a user (23) or injury may occur to the auditory meatus (24) or the tympanic membrane (25). While authorities vary on the external ear canal pressure (10) that may result in discomfort to a user (23) or injury to the auditory meatus (24) or the tympanic membrane (25), typically embodiments of the external ear canal pressure regulation device (1) would not be configured to operate in excess of about -50 kilopascals below the ambient pressure (11) or about +50 kilopascals above the ambient pressure (11).

(98) Now referring primarily to FIG. 2A, FIG. 5, and FIG. 11A through FIG. 11D, as to particular embodiments, the external ear canal pressure regulation device (1) can be configured to achieve an external ear canal pressure (10) which can be greater than the ambient pressure (11). As such, the valved conduit (5) can be coupled to the fluid flow generator (2) and the axial earpiece bore (4) of the earpiece (3) in the first configuration (45) to unidirectionally regulate the fluid flow (8) in the first direction (47) in the first fluid flow conduit (5) to egress from the axial earpiece bore (4) toward the external ear canal (6).

(99) As shown in FIG. 11A and FIG. 11B, the earpiece external surface (7) of the earpiece (3) can be sealably engaged with the external ear canal (6) as described above. Operation of the fluid flow generator (2) can compress an amount of fluid (74) in the first portion (53) of the first fluid flow conduit (49). As to those embodiments including a volumetrically adjustable element (13), the resiliently flexible wall (16) can be deformed to reduce the internal volume (19) of the volumetrically adjustable element (13) to compress the amount of fluid (74) in the first portion (53) of the first fluid flow conduit (49), resulting in a pressure differential (9) between opposite sides of the first valve (52) and the second valve (56). As to those embodiments including a piston (27) which reciprocally operates in a barrel (28), the piston (27) can travel within the barrel (28) to reduce the barrel internal volume, thus resulting in a pressure differential (9) between opposite sides of the first valve (52) and the second valve (56). Regardless of the configuration of the fluid flow generator (2), the pressure differential (9) between opposite sides of the first valve (52) can be such that a fluid pressure (75) in the first portion (53) of the first fluid flow conduit (49) can be greater than the fluid pressure (75) in the second portion (54) of the first fluid flow conduit (49). This pressure differential (9) can be sufficient to generate the open condition (69) of the first valve (52). The pressure differential (9) between opposite sides of the second valve (56) can be such that the fluid pressure (75) in the first portion (53) of the first fluid flow conduit (49) can be greater than the ambient pressure (11), which can be sufficient to generate the closed condition (68) of the second valve (56). The open condition (69) of the first valve (52) and the closed condition (68) of the second valve (56) can result in a fluid flow (8) in the first direction (47) in the first fluid flow conduit (49) which flows from the fluid flow generator (2) through the axial earpiece bore (4) to egress from the earpiece second end (29) into the external ear canal (6), thus increasing the external ear canal pressure (10) to greater than the ambient pressure (11).

(100) As shown in FIG. 11C, continued operation of the fluid flow generator (2) can further act to reduce the pressure differential (9) between opposite sides of the first valve (52) and the second valve (56), for example by allowing the resiliently flexible wall (16) of the volumetrically adjustable element (13) to return toward the non-deformed condition (21). The pressure differential (9) between opposite sides of the first valve (52) can be such that the fluid pressure (75) in the first portion (53) of the first fluid flow conduit (49) can be lesser than the fluid pressure (75) in the

second portion (54) of the first fluid flow conduit (49). This pressure differential (9) can be sufficient to generate the closed condition (68) of the first valve (52), thus interrupting the fluid flow (8) in the first fluid flow conduit (49) from the axial earpiece bore (4) toward the fluid flow generator (2) and thereby correspondingly maintaining the pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11).

(101) Again referring to FIG. 11C, continued reduction in fluid pressure (75) within the first portion (53) of the first fluid flow conduit (49) can result in a pressure differential (9) between opposite sides of the second valve (56) such that the fluid pressure (75) in the first portion (53) of the first fluid flow conduit (49) can be lesser than the ambient pressure (11), which can be sufficient to generate the open condition (69) of the second valve (56). As such, a fluid flow (8) can be generated from the ambient pressure (11) through the second fluid flow conduit (55) toward the fluid flow generator (2). As to those embodiments having a volumetrically adjustable element (13), the fluid flow (8) from the ambient pressure (11) toward the fluid flow generator (2) can allow the resiliently flexible wall (16) to return toward the non-deformed condition (21) by increasing the internal volume (19). As to those embodiments having a piston (27) which reciprocally operates in a barrel (28), the fluid flow (8) through the second fluid flow conduit (55) from the ambient pressure (11) toward the fluid flow generator (2) can allow the piston (27) to return to a location within the barrel (28) which increases the barrel internal volume.

(102) Now referring to FIG. 11D, the third valve (58) in the closed condition (68) can remain substantially leak-tight to fluid flow (8) in the third fluid flow conduit (57) up to about a 50 kilopascal pressure differential amplitude (59) between the external ear canal pressure (10) and the ambient pressure (11). Accordingly, the external ear canal pressure regulation device (1) can be operated to achieve a desired external ear canal pressure (10) greater than the ambient pressure (11) or to achieve a pre-selected external ear canal pressure (10) greater than the ambient pressure (11), beyond which results in the open condition (69) of the third valve (58), allowing a fluid flow (8) from the external ear canal pressure (10) toward ambient pressure (11) to maintain the desired or pre-selected external ear canal pressure (10).

(103) Now referring primarily to FIG. 5 and FIG. 11D, the third valve (58) can be operably coupled to a third valve pressure relief element (72) configured to allow manual operation of the third valve (58) between the closed condition (68) and the open condition (69), facilitating the return of the external ear canal pressure (10) toward the ambient pressure (11).

(104) Now referring primarily to FIG. 2B, FIG. 6, and FIG. 12A through FIG. 12D, as to particular embodiments, the external ear canal pressure regulation device (1) can be configured to achieve an external ear canal pressure (10) which can be lesser than the ambient pressure (11). As such, the valved conduit (5) can be coupled to the fluid flow generator (2) and the axial earpiece bore (4) of the earpiece (3) in the second configuration (46) to unidirectionally regulate the fluid flow (8) in the second direction (48) in the first fluid flow conduit (49) to ingress to the axial earpiece bore (4) from the external ear canal (6) toward the fluid flow generator (2).

(105) As shown in FIG. 12A and FIG. 12B, the earpiece external surface (7) of the earpiece (3) can be sealably engaged with the external ear canal (6) as described above. Operation of the fluid flow generator (2) can compress an amount of fluid (74) in the first portion (53) of the first fluid flow conduit (49). As to those embodiments including a volumetrically adjustable element (13), the resiliently flexible wall (16) can be deformed to reduce the internal volume (19) of the volumetrically adjustable element (13) to compress the amount of fluid (74) in the first portion (53) of the first fluid flow conduit (49), resulting in a pressure differential (9) between opposite sides of the first valve (52) and the second valve (56). As to those embodiments including a piston (27) which reciprocally operates in a barrel (28), the piston (27) can travel within the barrel (28) to reduce the barrel internal volume, thus resulting in a pressure differential (9) between opposite sides of the first valve (52) and the second valve (56). Regardless of the configuration of the fluid flow generator (2), the pressure differential (9) between opposite sides of the first valve (52) can be

such that a fluid pressure (75) in the first portion (53) of the first fluid flow conduit (49) can be greater than the fluid pressure (75) in the second portion (54) of the first fluid flow conduit (49). This pressure differential (9) can be sufficient to generate the closed condition (68) of the first valve (52). The pressure differential (9) between opposite sides of the second valve (56) can be such that the fluid pressure (75) in the first portion (53) of the first fluid flow conduit (49) can be greater than the ambient pressure (11), which can be sufficient to generate the open condition (69) of the second valve (56), which can generate a fluid flow (8) in the second fluid flow conduit (55) from the fluid flow generator (2) toward the ambient pressure (11).

(106) As shown in FIG. 12C, continued operation of the fluid flow generator (2) can further act to reduce the pressure differential (9) between opposite sides of the first valve (52) and the second valve (56), for example by allowing the resiliently flexible wall (16) of the volumetrically adjustable element (13) to return toward the non-deformed condition (21). The pressure differential (9) between opposite sides of the first valve (52) can be such that the fluid pressure (75) in the first portion (53) of the first fluid flow conduit (49) can be lesser than the fluid pressure (75) in the second portion (54) of the first fluid flow conduit (49). This pressure differential (9) can be sufficient to generate the open condition (69) of the first valve (52), which can result in a fluid flow (8) from the external ear canal (6) toward the fluid flow generator (2), thus decreasing the external ear canal pressure (10) to lesser than the ambient pressure (11).

(107) Again referring to FIG. 12C, continued reduction in fluid pressure (75) within the first portion (53) of the first fluid flow conduit (49) can result in a pressure differential (9) between opposite sides of the second valve (56) such that the fluid pressure (75) in the first portion (53) of the first fluid flow conduit (49) can be lesser than the ambient pressure (11), which can be sufficient to generate the closed condition (68) of the second valve (56), thus interrupting the fluid flow (8) in the second fluid flow conduit (55) from the fluid flow generator (2) toward the ambient pressure (11) and correspondingly maintaining the pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11).

(108) As to those embodiments having a volumetrically adjustable element (13), the fluid flow (8) from the external ear canal (6) toward the fluid flow generator (2) can allow the resiliently flexible wall (16) to return toward the non-deformed condition (21) by increasing the internal volume (19). As to those embodiments having a piston (27) which reciprocally operates in a barrel (28), the fluid flow (8) through the first fluid flow conduit (49) from the external ear canal (6) toward the fluid flow generator (2) can allow the piston (27) to return to a location within the barrel (28) which increases the barrel internal volume.

(109) Now referring to FIG. 12D, the third valve (58) in the closed condition (68) can remain substantially leak-tight to fluid flow (8) in the third fluid flow conduit (57) up to about a 50 kilopascal pressure differential amplitude (59) between the external ear canal pressure (10) and the ambient pressure (11). Accordingly, the external ear canal pressure regulation device (1) can be operated to achieve a desired external ear canal pressure (10) lesser than the ambient pressure (11) or to achieve a pre-selected external ear canal pressure (10) lesser than ambient pressure (11), beyond which results in the open condition (69) of the third valve (58), allowing a fluid flow (8) from the ambient pressure (11) toward the external ear canal (6) to maintain the desired or pre-selected external ear canal pressure (10).

(110) Now referring primarily to FIG. 6 and FIG. 12D, the third valve (58) can be operably coupled to a third valve pressure relief element (72) configured to allow manual operation of the third valve (58) between the closed condition (68) and the open condition (69), facilitating the return of the external ear canal pressure (10) toward the ambient pressure (11).

(111) Now referring primarily to FIG. 13A through FIG. 13F, a particular embodiment of the external ear canal pressure regulation device (1) can include a fluid flow generator (2) configured as a diaphragm (15), which has a resiliently flexible wall (16) having a wall external surface (17) and a wall internal surface (18), which defines an internal volume (19) (whether in whole or in part

as an assembly with the valved conduit (5)). The resiliently flexible wall (16) in a deformed condition (20) can decrease the internal volume (19), and in return toward a non-deformed condition (21), can increase the internal volume (19). The change in the internal volume (19) can generate a fluid flow (8) between the fluid flow generator (2) and an axial earpiece bore (4) of an earpiece (3), which can be regulated by a valved conduit (5).

(112) Again referring primarily to FIG. 13A through FIG. 13F, the external ear canal pressure regulation device (1) can include a valved conduit (5) fluidically coupled between the fluid flow generator (2) and the axial earpiece bore (4). As to particular embodiments, the earpiece (3) having the axial earpiece bore (4) and first, second, and third fluid flow conduits (49)(55)(57) can be included in a conduit body (41) having a configuration which removably couples to the fluid flow generator (2) and the first, second, and third valves (52)(56)(58). As to particular embodiments, the first and second valves (52)(56) can be included in a first valve assembly (83), which can be fluidically coupled between the fluid flow generator (2) and the first and second fluid flow conduits (49)(55) to interrupt the fluid flow (8) between the fluid flow generator (2) and the first and second fluid flow conduits (49)(55), thereby unidirectionally regulating the fluid flow (8) in the first and second fluid flow conduits (49)(55).

(113) The third valve (58) can be included in a second valve assembly (84), which can be fluidically coupled to a third fluid flow conduit (57) communicating between the first fluid flow conduit (49) and the ambient pressure (11) to interrupt the fluid flow (8) between the third fluid flow conduit (57) and the ambient pressure (11), thereby unidirectionally regulating the fluid flow (8) in the third fluid flow conduit (57) between the first fluid flow conduit (49) and the ambient pressure (11). As to particular embodiments, the fluid flow generator (2) and the first and second valve assemblies (83)(84) can be provided as a one-piece construct. As to other particular embodiments, the fluid flow generator (2), the first valve assembly (83), and the second valve assembly (84) can be provided as a plurality of pieces which can be assembled into a configuration capable of removably coupling to the conduit body (41).

(114) As to particular embodiments, the first and second valves (52)(56) included in the first valve assembly (83) can have any type of valve configuration, as described above, which can operate between a closed condition (68) and an open condition (69) to unidirectionally regulate fluid flow (8) in a pre-selected range. As an illustrative example, the first and second valves (52)(56) included in the first valve assembly (83) can both be configured as flapper valves (64) having hinges (94) (as shown in the example of FIG. 3B, and FIG. 13C though FIG. 13F) having opposing configurations.

(115) As to particular embodiments, the second valve assembly (84) can be operable to allow the external ear canal pressure (10) to return toward the ambient pressure (11). The second valve assembly (84) can include a resiliently deformable annular member (85) and a pressure relief element (70) configured as a deformation member (86) capable of deforming the resiliently deformable annular member (85). The resiliently deformable annular member (85) can have an annular member external surface (87) which can be disposed adjacent to and sealably engage with the conduit internal surface (88) of the first fluid flow conduit (49).

(116) An annular member aperture element (89) communicating between an annular member internal surface (90) and the annular member external surface (87) can align with the axial earpiece bore (4) to form a pass-through (91) between the first fluid flow conduit (49) and the axial earpiece bore (4). The deformation member (86) can be disposed through the third fluid flow conduit (57) communicating between the first fluid flow conduit (49) and the ambient pressure (11) such that a deformation member first end (92) can extend outward from the third fluid flow conduit (57) and the conduit body (41) and a deformation member second end (93) can deformably engage the annular member external surface (87). Upon gripping engagement of the deformation member first end (92), the deformation member (86) can be urged toward the resiliently deformable annular member (85) to deform the resiliently deformable annular member (85) such that the annular member external surface (87) disengages from the conduit internal surface (88) of the first fluid

flow conduit (49), thereby positioning the third valve (58) in the open condition (69) to allow the fluid flow (8) to flow between the axial earpiece bore (4), the first and third fluid flow conduits (49)(57), and the ambient pressure (11). As such, the third valve (58) in the open condition (69) can generate a fluid flow (8) between the external ear canal (6) and the ambient pressure (11) to return the external ear canal pressure (10) toward the ambient pressure (11), whether from an external ear canal pressure (10) greater than the ambient pressure (11) or an external ear canal pressure (10) lesser than the ambient pressure (11).

(117) The fluid flow generator (2), the first valve assembly (83), and the second valve assembly (84) can be removably coupled to the conduit body (41) in either one of a first configuration (45) or a second configuration (46). The releasably couplable surfaces (44) of the conduit body (41), the fluid flow generator (2), the first valve assembly (83), and the second valve assembly (84) can have sufficiently similar configurations to allow the conduit body (41) to removably couple to the fluid flow generator (2), the first valve assembly (83), and the second valve assembly (84) in either one of the first configuration (45) or the second configuration (46) depending upon whether the valved conduit (5) operates to achieve a pressure differential (9) having the external ear canal pressure (10) greater than the ambient pressure (11) (as shown in the example of FIG. 13E) or whether the valved conduit (5) operates to achieve a pressure differential (9) having the external ear canal pressure (10) lesser than ambient the pressure (11) (as shown in the example of FIG. 13F). As to particular embodiments, to removably couple, the releasably couplable surfaces (44) of the conduit body (41), the fluid flow generator (2), the first valve assembly (83), and the second valve assembly (84) can matably engage.

(118) As to the particular embodiments of the external ear canal pressure regulation device (1) described herein, the first and second valves (52)(56) fluidically couple to the corresponding first and second fluid flow conduits (49)(55) to interrupt the fluid flow (8) within the corresponding first and second fluid flow conduits (49)(55), whether the conduit body (41) can be positioned in the first configuration (45) or the second configuration (46).

(119) Now referring primarily to FIG. 13E, as an illustrative example, the fluid flow generator (2), the first valve assembly (83), and the conduit body (41) can be positioned in the first configuration (45) by removably coupling the fluid flow generator (2) and the first valve assembly (83) to the conduit body (41). In the first configuration (45), the first valve (52) can be operable to unidirectionally regulate the fluid flow (8) in the first fluid flow conduit (49), whereby the first valve (52) in the open condition (69) allows the fluid flow (8) to flow from the fluid flow generator (2) toward the axial earpiece bore (4) and, in the closed condition (68), precludes the fluid flow (8) from flowing between the fluid flow generator (2) and the axial earpiece bore (4). As such, the fluid flow (8), generated by deforming the resiliently flexible wall (16) of the diaphragm (15) to decrease the internal volume (19), can be regulated in a first direction (47) in the first fluid flow conduit (49) to egress from the axial earpiece bore (4) toward the external ear canal (6), thereby achieving a pressure differential (9) having the external ear canal pressure (10) greater than the ambient pressure (11). Upon achieving the desired pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11), the resiliently flexible wall (16) of the diaphragm (15) can be allowed to return to the non-deformed condition (21), thereby closing the first valve (52) and precluding the fluid flow (8) from flowing from the axial earpiece bore (4) toward the fluid flow generator (2), thereby maintaining the desired pressure differential (9).

(120) In the first configuration (45), the second valve (56) can be operable to unidirectionally regulate the fluid flow (8) in the second fluid flow conduit (55), whereby the second valve (56) in the open condition (69) allows the fluid flow (8) to flow from the ambient pressure (11) toward the fluid flow generator (2) and, in the closed condition (68), precludes the fluid flow (8) from flowing between the ambient pressure (11) and the fluid flow generator (2).

(121) As such, the fluid flow (8), generated upon return of the resiliently flexible wall (16) of the diaphragm (15) to the non-deformed condition (21) to increase the internal volume (19), can be

regulated in the second fluid flow conduit (55) to ingress from the ambient pressure (11) toward the fluid flow generator (2), thereby maintaining the desired pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11) while allowing the resiliently flexible wall (16) of the diaphragm (15) to return toward the non-deformed condition (21).

(122) As to particular embodiments, the fluid flow generator (2), the first valve assembly (83), and the conduit body (41) in the first configuration (45) can further include the second valve assembly (84), positioned such that the third valve (58) can be fluidically coupled to the third fluid flow conduit (57) communicating between the first fluid flow conduit (49) and the ambient pressure (11) to unidirectionally regulate the fluid flow (8) in the third fluid flow conduit (57) between the first fluid flow conduit (49) and the ambient pressure (11). The third valve (58) in the open condition (69) allows the fluid flow (8) to flow from the axial earpiece bore (4), through the first and third fluid flow conduits (49)(57), and toward the ambient pressure (11) and, in the closed condition (68), precludes the fluid flow (8) from flowing between the axial earpiece bore (4) and the ambient pressure (11). As such, the third valve (58) in the open condition (69) can generate a fluid flow (8) from the external ear canal (6) toward the ambient pressure (11) to return the external ear canal pressure (10) toward the ambient pressure (11).

(123) As to particular embodiments, the fluid flow generator (2), the first valve assembly (83), and the second valve assembly (84) can be removed from the conduit body (41) to disassemble the first configuration (45). The first valve assembly (83) can be rotated without any structural alteration to reverse orientation of the first valve assembly (83) in relation to the conduit body (41), thereby achieving the second configuration (46).

(124) Now referring primarily to FIG. 13F, as an illustrative example, the fluid flow generator (2), the first valve assembly (83), and the conduit body (41) can be positioned in the second configuration (46) as described above. In the second configuration (46), the second valve (56) can be operable to unidirectionally regulate the fluid flow (8) in the second fluid flow conduit (55), whereby the second valve (56) in the open condition (69) allows the fluid flow (8) to flow from the fluid flow generator (2) toward the ambient pressure (11) and, in the closed condition (68), precludes the fluid flow (8) from flowing between the fluid flow generator (4) and the ambient pressure (11). As such, the fluid flow (8), generated by deforming the resiliently flexible wall (16) of the diaphragm (15) to decrease the internal volume (19), can be regulated in the second fluid flow conduit (55) to egress from the second fluid flow conduit (55) toward the ambient pressure (11). The resiliently flexible wall (16) of the diaphragm (15) can be allowed to return to the non-deformed condition (21), thereby closing the second valve (56) and precluding the fluid flow (8) from flowing between the fluid flow generator (2) and the ambient pressure (11).

(125) In the second configuration (46), the first valve (52) can be operable to unidirectionally regulate the fluid flow (8) in the first fluid flow conduit (49), whereby the first valve (52) in the open condition (69) allows the fluid flow (8) to flow from the axial earpiece bore (4) toward the fluid flow generator (2) and, in the closed condition (68), precludes the fluid flow (8) from flowing between the axial earpiece bore (4) and the fluid flow generator (2).

(126) As such, the fluid flow (8), generated upon return of the resiliently flexible wall (16) of the diaphragm (15) to the non-deformed condition (21) to increase the internal volume (19), can be regulated in a second direction (48) in the first fluid flow conduit (56) to ingress from the external ear canal (6) to the axial earpiece bore (4) toward the fluid flow generator (2), thereby achieving a pressure differential (9) having the external ear canal pressure (10) lesser than the ambient pressure (11). Upon achieving the desired pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11), the first valve (52) can return toward the closed condition (68), precluding the fluid flow (8) from flowing between the axial earpiece bore (4) and the fluid flow generator (2), thereby maintaining the desired pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11).

(127) As to particular embodiments, the fluid flow generator (2), the first valve assembly (83), and

the conduit body (41) in the second configuration (46) can further include the second valve assembly (84), positioned such that the third valve (58) can be fluidically coupled to the third fluid flow conduit (57) communicating between the first fluid flow conduit (49) and the ambient pressure (11) to unidirectionally regulate the fluid flow (8) in the third fluid flow conduit (57) between the first fluid flow conduit (49) and the ambient pressure (11). The third valve (58) in the open condition (69) allows the fluid flow (8) to flow from the ambient pressure (11), through the third and first fluid flow conduits (57)(49), and toward the axial earpiece bore (4) and, in the closed condition (68), precludes the fluid flow (8) from flowing between the ambient pressure (11) and the axial earpiece bore (4). As such, the third valve (58) in the open condition (69) can generate a fluid flow (8) from the ambient pressure (11) toward the external ear canal (6) to return the external ear canal pressure (10) toward the ambient pressure (11).

(128) Now referring primarily to FIG. 14A through FIG. 15G, which show a pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11) achieved over a time period (76) by operation of embodiments of the external ear canal pressure regulation device (1). As to particular embodiments, a fluid flow generator (2) comprising a volumetrically adjustable element (13) can be operated from a non-deformed condition (21) toward a deformed condition (20) to generate a fluid flow (8) in a first direction (47) which egresses from the axial earpiece bore (4) toward the external ear canal (6) over a time period (76), resulting in a positive external ear canal pressure (10) relative to the ambient pressure (11) (as shown in the examples of FIG. 14A through FIG. 14G). As to other particular embodiments, a fluid flow generator (2) comprising a volumetrically adjustable element (13) can be operated from a deformed condition (20) toward a non-deformed condition (21) to generate a fluid flow (8) in a second direction (48) which ingresses to the axial earpiece bore (4) from the external ear canal (6) toward the fluid flow generator (2) over a time period (76), resulting in a negative external ear canal pressure (10) relative to the ambient pressure (11) (as shown in the examples of FIG. 15A through FIG. 15G).

(129) Now referring primarily to FIG. 14A and FIG. 15A, the fluid flow generator (2) can be operated to maintain a constant pressure amplitude (77) over a time period (76). As to particular embodiments, a constant pressure amplitude (77) can be maintained substantially without fluid flow (8) of a fluid volume (12) within the external ear canal (6) over the time period (76). As an illustrative example, the external ear canal pressure regulation device (1) having the earpiece external surface (7) sealably engaged with the external ear canal (6), as described above, can be operated by manipulating the fluid flow generator (2) to generate a fluid flow (8) of a fluid volume (12) between the fluid flow generator (2) and the external ear canal (6) through the axial earpiece bore (4) of the earpiece (3) to achieve a pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11). Once the desired pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11) has been achieved, the pressure amplitude (77) can be maintained for a time period (76) without additional fluid flow (8) of the fluid volume (12), for example by operation of the first valve (52) configured to allow unidirectional fluid flow (8) of the fluid volume (12) through the first fluid flow conduit (49) such that the fluid volume (12) can only either ingress or egress from the external ear canal (6). As to other embodiments, once the desired pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11) has been achieved, the pressure amplitude (77) can be maintained for a time period (76) by additional fluid flow (8) of a fluid volume (12) to or from the external ear canal (6) to offset leakage about engagement of the earpiece external surface (7) with the external ear canal (6). As to other embodiments, the pressure amplitude (77) can be maintained for a time period (76) by continuous fluid flow (8) of a fluid volume (12) in the external ear canal (6).

(130) The constant pressure amplitude (77) can be maintained over a time period (76) to alleviate a disorder symptom or treat a disorder, whereby the constant pressure amplitude (77) can be in a range of between about +50 kilopascals above the ambient pressure (11) to about -50 kilopascals

below the ambient pressure (11). A positive external ear canal pressure (10) relative to the ambient pressure (11) can be achieved by maintaining the constant pressure amplitude (77) in a range of between about 0 kilopascals to about +50 kilopascals above the ambient pressure (11).

Alternatively, a negative external ear canal pressure (10) relative to the ambient pressure (11) can be achieved by maintaining the constant pressure amplitude (77) in a range of between about -50 kilopascals to about 0 kilopascals below the ambient pressure (11).

(131) Now referring primarily to FIG. 14B through FIG. 14G and FIG. 15B through FIG. 15G, the fluid flow generator (2) can be configured for repeated operation to generate a fluid flow (8) having a pressure wave (78) including a pressure wave amplitude (77) and a pressure wave frequency (79). As to particular embodiments, a fluid flow generator (2) comprising a volumetrically adjustable element (13) can be operated from a non-deformed condition (21) toward a deformed condition (20) to generate a fluid flow (8) which egresses from the axial earpiece bore (4) toward the external ear canal (6) over a time period (76), whereby the fluid flow (8) has a pressure wave (78) including a pressure wave amplitude (77) and a pressure wave frequency (79) which results in a positive external ear canal pressure (10) relative to the ambient pressure (11) (as shown in the examples of FIG. 14B through FIG. 14G). As to other particular embodiments, a fluid flow generator (2) comprising a volumetrically adjustable element (13) can be operated from a deformed condition (20) toward a non-deformed condition (21) to generate a fluid flow (8) which ingresses to the axial earpiece bore (4) from the external ear canal (6) toward the fluid flow generator (2) over a time period (76), whereby the fluid flow (8) has a pressure wave (78) including a pressure wave amplitude (77) and a pressure wave frequency (79) which results in a negative external ear canal pressure (10) relative to the ambient pressure (11) (as shown in the examples of FIG. 15B through FIG. 15G).

(132) Again referring primarily to FIG. 14A through FIG. 15G, the fluid flow generator (2) can be configured for repeated operation to generate a fluid flow (8) having a pressure wave (78) including a pressure wave amplitude (77) as described above and a pressure wave frequency (79) typically in a range of between 0 Hertz to about 10 Hertz; however, embodiments can have a lesser or greater pressure wave frequency (79) depending upon the application. As to particular embodiments, one or more pressure wave frequencies (79) can be selected from the group including or consisting of: between 0 Hertz to about 1 Hertz, between about 0.5 Hertz to about 1.5 Hertz, between about 1 Hertz to about 2 Hertz, between about 1.5 Hertz to about 2.5 Hertz, between about 2 Hertz to about 3 Hertz, between about 2.5 Hertz to about 3.5 Hertz, between about 3 Hertz to about 4 Hertz, between about 3.5 Hertz to about 4.5 Hertz, between about 4 Hertz to about 5 Hertz, between about 4.5 Hertz to about 5.5 Hertz, between about 5 Hertz to about 6 Hertz, between about 5.5 Hertz to about 6.5 Hertz, between about 6 Hertz to about 7 Hertz, between about 6.5 Hertz to about 7.5 Hertz, between about 7 Hertz to about 8 Hertz, between about 7.5 Hertz to about 8.5 Hertz, between about 8 Hertz to about 9 Hertz, between about 8.5 Hertz to about 9.5 Hertz, and between about 9 Hertz to about 10 Hertz.

(133) One or more pressure wave frequencies (79) can be generated with the external ear canal pressure regulation device (1) depending upon the method of use, which can be further influenced by factors such as user (23) anatomy, physiology, or biochemistry of the auditory meatus (24); disorder symptom targeted for alleviation; disorder targeted for treatment; observable effect(s) of using one or more pressure wave frequencies (79) in a particular method of using the external ear canal pressure regulation device (1); or the like; or combinations thereof; but not so much as to cause discomfort to the user (23) or injury to the auditory meatus (24) or the tympanic membrane (25).

(134) The pressure wave (78) can oscillate with a desired pressure wave frequency (79) within only positive pressure amplitudes (77) in a range of between 0 kilopascals to about +50 kilopascals above the ambient pressure (11) (as shown in the examples of FIG. 14B through FIG. 14G), which can correspondingly generate a positive external ear canal pressure (10) relative to the ambient

pressure (11) by increasing the external ear canal pressure (10) relative to the ambient pressure (11), for example to alleviate a disorder symptom or treat a disorder. As to yet other particular embodiments, the pressure wave (78) can oscillate with a desired pressure wave frequency (79) within only negative pressure amplitudes (77) in a range of between about -50 kilopascals to 0 kilopascals below the ambient pressure (11) (as shown in the examples of FIG. 15B through FIG. 15G), which can correspondingly generate a negative external ear canal pressure (10) relative to the ambient pressure (11) by decreasing the external ear canal pressure (10) relative to the ambient pressure (11), for example to alleviate a disorder symptom or treat a disorder.

(135) Again referring primarily to FIG. 14B through FIG. 15G, the pressure wave (78) can have a numerous and wide variety of waveforms, depending upon the application, corresponding to the numerous and wide variety of symptoms which can be alleviated or disorders which can be treated by operation of the external ear canal pressure regulation device (1). As illustrative examples, the pressure wave (78) can be sine wave having smooth repetitive periodic oscillations (as shown in the example of FIG. 14B and FIG. 15B), a square wave in which the pressure amplitude (77) alternates at a steady frequency between fixed minimum and maximum values, a rectangular wave, a trapezoidal wave or a truncated wave in which the apex of the pressure wave (78) has a constant pressure amplitude (77) over a time period (76) (as shown in the example of FIG. 14C, FIG. 15C, FIG. 14F, and FIG. 15F), a triangle wave having linear leading and trailing edges (as shown in the example of FIG. 14D and FIG. 15D), a sawtooth wave in which the leading edge changes pressure amplitude (77) over a time period (76) which is greater than the time period (76) in which the trailing edge changes pressure amplitude (77) (as shown in the example of FIG. 14E), a reverse sawtooth wave in which the leading edge changes pressure amplitude (77) over a time period (76) which is lesser than the time period (76) in which the trailing edge changes pressure amplitude (77) (as shown in the example of FIG. 15E), or combinations thereof (as shown in the example of FIG. 14G and FIG. 15G).

(136) As to particular embodiments, the fluid flow generator (2) and the third valve pressure relief element (72) coupled to the third valve (58) can be alternately repeatedly operated to generate a fluid flow (8) having a pressure wave (78) including a pressure wave amplitude (77) and a pressure wave frequency (79).

(137) Now referring primarily to FIG. 14A through FIG. 14G, which show a pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11) achieved over a time period (76) by operation of embodiments of the external ear canal pressure regulation device (1) to generate a positive external ear canal pressure (10) relative to the ambient pressure (11) (as shown in the examples of FIG. 2A, FIG. 5, and FIG. 11A through FIG. 11D). The particular embodiment of the external ear canal pressure regulation device (1) includes a third valve (58) which remains substantially leak-tight to fluid flow (8) in the third fluid flow conduit (57) up to about a 50 kilopascal pressure differential amplitude (59) between the external ear canal pressure (10) and the ambient pressure (11) (represented in each graph as having a pressure amplitude (77) of zero). However, this need not limit embodiments solely to those capable of generating a positive external ear canal pressure (10) relative to the ambient pressure (11) pre-selected to a maximum of +50 kilopascals. Other embodiments can operate to generate a positive external ear canal pressure (10) relative to the ambient pressure (11) which can be any amount greater than the ambient pressure (11), but not an amount so great as to cause discomfort to the user (23) or injury to the auditory meatus (24) or the tympanic membrane (25).

(138) Now referring primarily to FIG. 14A, as to a particular embodiment, the fluid flow generator (2) can be operated to generate an external ear canal pressure (10) having a maximum pressure amplitude (77) of up to about +50 kilopascals above the ambient pressure (11). The external ear canal pressure (10) can be maintained for a time period (76) at about +50 kilopascals above the ambient pressure (11). After elapse of the time period (76), the third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11).

The operation can be, but is not necessarily, repeated.

(139) Now referring primarily to FIG. 14B, as to a particular embodiment, the fluid flow generator (2) can be operated to generate an external ear canal pressure (10) having a maximum pressure amplitude (77) of up to about +50 kilopascals above the ambient pressure (11). The third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11), whereby the pressure wave (78) can be a sine wave having smooth repetitive periodic oscillations. The operation can be repeated to administer a pulsatile change in the external ear canal pressure (10).

(140) Now referring primarily to FIG. 14D, as to a particular embodiment, the fluid flow generator (2) can be operated to generate an external ear canal pressure (10) having a maximum pressure amplitude (77) of up to about +50 kilopascals above the ambient pressure (11). The third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11), whereby the pressure wave (78) can be a triangle wave having linear leading and trailing edges. The operation can be repeated to administer a pulsatile change in the external ear canal pressure (10).

(141) Now referring primarily to FIG. 14E, as to a particular embodiment, the fluid flow generator (2) can be operated to generate an external ear canal pressure (10) having a maximum pressure amplitude (77) of up to about +50 kilopascals above the ambient pressure (11). The third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11), whereby the pressure wave (78) can be a sawtooth wave in which the leading edge changes pressure amplitude (77) over a time period (76) which is greater than the time period (76) in which the trailing edge changes pressure amplitude (77). The operation can be repeated to administer a pulsatile change in the external ear canal pressure (10).

(142) Now referring primarily to FIG. 14C and FIG. 14F, as to a particular embodiment, the fluid flow generator (2) can be operated to generate an external ear canal pressure (10) having a maximum pressure amplitude (77) of up to about +50 kilopascals above the ambient pressure (11). The external ear canal pressure (10) can be maintained for a time period (76) at about +50 kilopascals above the ambient pressure (11). After elapse of the time period (76), the third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11). The operation can be repeated to administer a pulsatile change in the external ear canal pressure (10).

(143) Now referring primarily to FIG. 14G, as to a particular embodiment, the fluid flow generator (2) can be operated to increase the external ear canal pressure (10) in a series of incremental pressure increases having a maximum pressure amplitude (77) of about +50 kilopascals above the ambient pressure (11). Each of the series of incremental pressure increases can increase the external ear canal pressure (10) by about 10 kilopascals to about 15 kilopascals above the ambient pressure (11), with each incremental pressure increase in the external ear canal pressure (10) maintained for a time period (76). After achieving the maximum pressure amplitude (77) of about +50 kilopascals above the ambient pressure (11) and elapse of the time period (76), the third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11). The operation can be repeated to administer a pulsatile change in the external ear canal pressure (10).

(144) Now referring primarily to FIG. 15A through FIG. 15G, which show a pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11) achieved over a time period (76) by operation of embodiments of the external ear canal pressure regulation device (1) to generate a negative external ear canal pressure (10) relative to the ambient pressure (9) (as shown in the examples of FIG. 2B, FIG. 6, and FIG. 12A through FIG. 12D). The particular embodiment of the external ear canal pressure regulation device (1) includes a third valve (58) which remains substantially leak-tight to fluid flow (8) in the third fluid flow conduit (57) up to about a 50 kilopascal pressure differential amplitude (59) between the external ear canal pressure (10) and the

ambient pressure (11) (represented in each graph as having a pressure amplitude (77) of zero). However, this need not limit embodiments solely to those capable of generating a negative external ear canal pressure (10) relative to the ambient pressure (77) pre-selected to a maximum of -50 kilopascals. Other embodiments can operate to generate a negative external ear canal pressure (10) relative to the ambient pressure (11) which can be any amount lesser than the ambient pressure (11), but not an amount so great as to cause discomfort to the user (23) or injury to the auditory meatus (24) or the tympanic membrane (25).

(145) Now referring primarily to FIG. 15A, as to a particular embodiment, the fluid flow generator (2) can be operated to generate an external ear canal pressure (10) having a maximum pressure amplitude (77) of up to about -50 kilopascals below the ambient pressure (11). The external ear canal pressure (10) can be maintained for a time period (76) at about -50 kilopascals below the ambient pressure (11). After elapse of the time period (76), the third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11). The operation can be, but is not necessarily, repeated.

(146) Now referring primarily to FIG. 15B, as to a particular embodiment, the fluid flow generator (2) can be operated to generate an external ear canal pressure (10) having a maximum pressure amplitude (77) of up to about -50 kilopascals below the ambient pressure (11). The third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11), whereby the pressure wave (78) can be a sine wave having smooth repetitive periodic oscillations. The operation can be repeated to administer a pulsatile change in the external ear canal pressure (10).

(147) Now referring primarily to FIG. 15D, as to a particular embodiment, the fluid flow generator (2) can be operated to generate an external ear canal pressure (10) having a maximum pressure amplitude (77) of up to about -50 kilopascals below the ambient pressure (11). The third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11), whereby the pressure wave (78) can be a triangle wave having linear leading and trailing edges. The operation can be repeated to administer a pulsatile change in the external ear canal pressure (10).

(148) Now referring primarily to FIG. 15E, as to a particular embodiment, the fluid flow generator (2) can be operated to generate an external ear canal pressure (10) having a maximum pressure amplitude (77) of up to about -50 kilopascals below the ambient pressure (11). The third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11), whereby the pressure wave (78) can be a reverse sawtooth wave in which the leading edge changes pressure amplitude (77) over a time period (76) which is lesser than the time period (76) in which the trailing edge changes pressure amplitude (77). The operation can be repeated to administer a pulsatile change in the external ear canal pressure (10).

(149) Now referring primarily to FIG. 15C and FIG. 15F, as to a particular embodiment, the fluid flow generator (2) can be operated to generate an external ear canal pressure (10) having a maximum pressure amplitude (77) of up to about -50 kilopascals below the ambient pressure (11). The external ear canal pressure (10) can be maintained for a time period (76) at about -50 kilopascals below the ambient pressure (11). After elapse of the time period (76), the third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11). The operation can be repeated to administer a pulsatile change in the external ear canal pressure (10).

(150) Now referring primarily to FIG. 15G, as to a particular embodiment, the fluid flow generator (2) can be operated to decrease the external ear canal pressure (10) in a series of incremental pressure decreases having a maximum pressure amplitude (77) of about -50 kilopascals below the ambient pressure (11). Each of the series of incremental pressure decreases can decrease the external ear canal pressure (10) by about -10 kilopascals to about -15 kilopascals below the ambient pressure (11), with each incremental pressure decrease in the external ear canal pressure

(10) maintained for a time period (76). After achieving the maximum pressure amplitude (77) of about -50 kilopascals below the ambient pressure (11) and elapse of the time period (76), the third valve pressure relief element (72) can be operated to return the external ear canal pressure (10) toward the ambient pressure (11). The operation can be repeated to administer a pulsatile change in the external ear canal pressure (10).

(151) As to particular embodiments, the external ear canal pressure regulation device (1) can further include a housing (80) having a housing internal surface defining a generally hollow internal space in which components of the external ear canal pressure regulation device (1) can be housed. As to particular embodiments, the housing (80) can be configured to fill a concha area (82) of the ear (35), whether in whole or in part, without extending any substantial distance outside of the external auditory meatus (24), thereby providing a discrete, unobtrusive, portable configuration which can be used upon occurrence to alleviate one or more disorder symptoms, for example neurologically-mediated pain, or treat one or more disorders, for example craniofacial pain syndromes or headache syndromes.

(152) While the fluid flow generator (2) of the external ear canal pressure regulation device (1) above described typically delivers a fluid flow (8) of air to the external ear canal (6) to achieve the pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11), this is not intended to be limiting with respect to the wide variety of fluids which can be delivered to the external ear canal (6) by the external ear canal pressure regulation device (1). As illustrative examples, the wide variety of fluids can include: a purified gas, such as oxygen, nitrogen, argon, or the like; a mixture of partial pressures of gases; a liquid, such as water, oil, alcohol, or the like; or combinations thereof.

(153) Additionally, while the fluid flow (8) or the transfer of a fluid volume (12) between components of the external ear canal pressure regulation device (1), between components of the external ear canal pressure regulation device (1) and the external ear canal (6), or between components of the external ear canal pressure regulation device (1) and the ambient pressure (11) can be as above described as typically between a first point and a second point for the purpose of brevity, the fluid flow (8) or the transfer of the fluid volume (12) includes all points within the manifold fluid flow path (62) between the first point and the second point. For example, a fluid volume (12) transferred from the fluid flow generator (2) to the external ear canal (6) can travel along a fluid flow path (62) including the fluid flow generator (2), the first fluid flow conduit first end (50), the first portion (53) of the first fluid flow conduit (49), the first valve (52), the second portion (54) of the first fluid flow conduit (49), the first fluid flow conduit second end (51), the earpiece first end (31), the axial earpiece bore (4), the earpiece second end (29), and the external ear canal (6).

(154) A method of producing particular embodiments of the external ear canal pressure regulation device (1) can include providing a fluid flow generator (2) capable of generating a fluid flow (8); providing a valved conduit (5) capable of fluidically coupling to the fluid flow generator (2), the valved conduit (5) having a first fluid flow conduit (49); providing a first valve (52) capable of interrupting the first fluid flow conduit (49) to unidirectionally regulate the fluid flow (8) in the first fluid flow conduit (49); and providing an axial earpiece bore (4), which communicates between an earpiece first end (31) and an earpiece second end (32) of an earpiece (3), the axial earpiece bore (4) capable of fluidically coupling to the valved conduit (5) opposite the fluid flow generator (2), the earpiece (3) having a compliant earpiece external surface (7) configured to sealably engage an external ear canal (6) as a barrier between an external ear canal pressure (10) and an ambient pressure (11).

(155) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing a fluid flow generator (2) having a configuration capable of generating the fluid flow (8) having a fluid volume (12) in a range of between 0 milliliters to about 20 milliliters. As to particular embodiments, the fluid volume (12) can have a

pre-selected fluid volume (12), which can be selected from one or more of the group including or consisting of: between 0 milliliters to about 2 milliliters, between about 1 milliliter to about 3 milliliters, between about 2 milliliters to about 4 milliliters, between about 3 milliliters to about 5 milliliters, between about 4 milliliters to about 6 milliliters, between about 5 milliliters to about 7 milliliters, between about 6 milliliters to about 8 milliliters, between about 7 milliliters to about 9 milliliters, between about 8 milliliters to about 10 milliliters, between about 9 milliliters to about 11 milliliters, between about 10 milliliters to about 12 milliliters, between about 11 milliliters to about 13 milliliters, between about 12 milliliters to about 14 milliliters, between about 13 milliliters to about 15 milliliters, between about 14 milliliters to about 16 milliliters, between about 15 milliliters to about 17 milliliters, between about 16 milliliters to about 18 milliliters, between about 17 milliliters to about 19 milliliters, and between about 18 milliliters to about 20 milliliters.

(156) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing the valved conduit (5) having a configuration capable of removably coupling to the fluid flow generator (2) and the earpiece (3). As to particular embodiments, the method can further include providing the valved conduit (5) having a configuration capable of coupling in a first configuration (45) with the fluid flow generator (2) and the earpiece (3) to unidirectionally regulate the fluid flow (8) in a first direction (47) in the first fluid flow conduit (49). As to other particular embodiments, the method can further include providing the valved conduit (5) having a configuration capable of coupling in a second configuration (46) with the fluid flow generator (2) and the earpiece (3) to unidirectionally regulate the fluid flow (8) in a second direction (48) in the first fluid flow conduit (49).

(157) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing the first valve (52) having a configuration capable of dividing the first fluid flow conduit (49) into a first portion (53) proximate a first fluid flow conduit first end (50) and a second portion (54) proximate a first fluid flow conduit second end (51), and further comprising providing a second fluid flow conduit (55) having a configuration capable of fluidically coupling between the first portion (53) of the first fluid flow conduit (49) and the ambient pressure (11), and further comprising providing a second valve (56) having a configuration capable of interrupting the second fluid flow conduit (55) to unidirectionally regulate the fluid flow (8) in the second fluid flow conduit (55).

(158) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing the fluid flow generator (2) configured as a volumetrically adjustable element (13) having an internal volume (19), the volumetrically adjustable element (13) having a deformed condition (20) which decreases the internal volume (19) to generate the fluid flow (8) in the first fluid flow conduit (49), the first valve (52) unidirectionally regulating the fluid flow (8) to egress from the axial earpiece bore (4) of the earpiece (3). As to particular embodiments, the volumetrically adjustable element (13) can return to a non-deformed condition (21) which increases the internal volume (19) to generate the fluid flow (8) in the second fluid flow conduit (55), the second valve (56) unidirectionally regulating the fluid flow (8) to ingress from the ambient pressure (11) toward the volumetrically adjustable element (13), the first valve (52) interrupting the fluid flow (8) in the first fluid flow conduit (49) from the second portion (54) toward the first portion (53).

(159) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing the fluid flow generator (2) configured as a volumetrically adjustable element (13) having an internal volume (19), the volumetrically adjustable element (13) having a deformed condition (20) which decreases the internal volume (19) to generate the fluid flow (8) in the second fluid flow conduit (55), the second valve (52) unidirectionally regulating the fluid flow (8) to egress from the second fluid flow conduit (55) toward the ambient pressure (11). As to particular embodiments, the volumetrically adjustable element (13) can return to a non-deformed condition (21) which increases the internal volume (19)

to generate the fluid flow (8) in the first fluid flow conduit (49), the first valve (52) unidirectionally regulating the fluid flow (8) to ingress from the axial earpiece bore (4) of the earpiece (3) toward the volumetrically adjustable element (13), the second valve (56) interrupting the fluid flow (8) in the second fluid flow conduit (55) from the ambient pressure (11) toward the first portion (53).

(160) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing a third fluid flow conduit (57) having a configuration capable of fluidically coupling between the second portion (54) of the first fluid flow conduit (49) and the ambient pressure (11), and further comprising providing a third valve (58) having a configuration capable of interrupting the third fluid flow conduit (57) to unidirectionally regulate the fluid flow (8) in the third fluid flow conduit (57). As to particular embodiments, the method can further include providing the third valve (58) having a configuration capable of regulating the fluid flow (8) to egress to the ambient pressure (11). As to other particular embodiments, the method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing the third valve (58) having a configuration capable of regulating the fluid flow (8) to ingress from the ambient pressure (11).

(161) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing the third valve (58) having a configuration capable of interrupting the fluid flow (8) between the second portion (54) of the first fluid flow conduit (49) and the ambient pressure (11) until a pressure differential (9) between the second portion (54) of the first fluid conduit (49) and the ambient pressure (11) exceeds a pre-selected pressure differential (9) having a pressure differential amplitude (59) of between 0 kilopascals to about 50 kilopascals. As to particular embodiments, the one or more pre-selected pressure differential amplitudes (59) can be selected from the group including or consisting of: between 0 kilopascals to about 5 kilopascals, between about 2.5 kilopascals to about 7.5 kilopascals, between about 5 kilopascals to about 10 kilopascals, between about 7.5 kilopascals to about 12.5 kilopascals, between about 10 kilopascals to about 15 kilopascals, between about 12.5 kilopascals to about 17.5 kilopascals, between about 15 kilopascals to about 20 kilopascals, between about 17.5 kilopascals to about 22.5 kilopascals, between about 20 kilopascals to about 25 kilopascals, between about 22.5 kilopascals to about 27.5 kilopascals, between about 25 kilopascals to about 30 kilopascals, between about 27.5 kilopascals to about 32.5 kilopascals, between about 30 kilopascals to about 35 kilopascals, between about 32.5 kilopascals to about 37.5 kilopascals, between about 35 kilopascals to about 40 kilopascals, between about 37.5 kilopascals to about 42.5 kilopascals, between about 40 kilopascals to about 45 kilopascals, between about 42.5 kilopascals to about 47.5 kilopascals, and between about 45 kilopascals to about 50 kilopascals.

(162) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing a second valve pressure relief element (71) having a configuration capable of coupling to the second valve (56) and a third valve pressure relief element (72) having a configuration capable of coupling to the third valve (58), each one manually operable to correspondingly generate the fluid flow (8) in the second or third fluid flow conduits (55)(57).

(163) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing the valved conduit (5) having a configuration capable of coupling to the fluid flow generator (2) and the earpiece (3) in a first configuration (45) to unidirectionally regulate the fluid flow (8) in a first direction (47) in the first fluid flow conduit (49) to egress from the axial earpiece bore (4) of the earpiece (3). As to particular embodiments, the method can further include providing the fluid flow generator (2) configured as a volumetrically adjustable element (13) operable from a non-deformed condition (21) toward a deformed condition (20) to generate the fluid flow (2) which egresses from the axial earpiece bore (4) over a time period (76).

(164) The method of producing particular embodiments of the external ear canal pressure

regulation device (1) can further include providing the fluid flow generator (2) having a configuration capable of repeated operation from the non-deformed condition (21) toward the deformed condition (20) to generate the fluid flow (8) having a pressure wave (78) including a pressure wave amplitude (77) and a pressure wave frequency (79). As to particular embodiments, the pressure wave frequency (79) can be in a range of between 0 Hertz to about 10 Hertz. As to particular embodiments, one or more pressure wave frequencies (79) can be selected from the group including or consisting of: between 0 Hertz to about 1 Hertz, between about 0.5 Hertz to about 1.5 Hertz, between about 1 Hertz to about 2 Hertz, between about 1.5 Hertz to about 2.5 Hertz, between about 2 Hertz to about 3 Hertz, between about 2.5 Hertz to about 3.5 Hertz, between about 3 Hertz to about 4 Hertz, between about 3.5 Hertz to about 4.5 Hertz, between about 4 Hertz to about 5 Hertz, between about 4.5 Hertz to about 5.5 Hertz, between about 5 Hertz to about 6 Hertz, between about 5.5 Hertz to about 6.5 Hertz, between about 6 Hertz to about 7 Hertz, between about 6.5 Hertz to about 7.5 Hertz, between about 7 Hertz to about 8 Hertz, between about 7.5 Hertz to about 8.5 Hertz, between about 8 Hertz to about 9 Hertz, between about 8.5 Hertz to about 9.5 Hertz, and between about 9 Hertz to about 10 Hertz.

(165) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing a third valve pressure relief element (72) having a configuration capable of coupling to the third valve (56), whereby the fluid flow generator (2) and the third valve pressure relief element (72) can be capable of alternate repeated operation to generate the fluid flow (2) having the pressure wave (78) including the pressure wave amplitude (77) and the pressure wave frequency (79).

(166) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing the valved conduit (5) having a configuration capable of coupling to the fluid flow generator (2) and the earpiece (3) in a second configuration (46) to unidirectionally regulate the fluid flow (8) in a second direction (48) in the first fluid flow conduit (49) to ingress to the axial earpiece bore (4) of the earpiece (3). As to particular embodiments, the method can further include providing the fluid flow generator (2) configured as a volumetrically adjustable element (13) operable from a deformed condition (20) toward a non-deformed condition (21) to generate the fluid flow (2) which ingresses from the axial earpiece bore (4) over a time period (76).

(167) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing the fluid flow generator (2) having a configuration capable of repeated operation from the deformed condition (20) toward the non-deformed condition (21) to generate the fluid flow (8) having a pressure wave (78) including a pressure wave amplitude (77) and a pressure wave frequency (79). As to particular embodiments, the pressure wave frequency (79) can be in a range of between 0 Hertz to about 10 Hertz. As to particular embodiments, one or more pressure wave frequencies (79) can be selected from the group including or consisting of: between 0 Hertz to about 1 Hertz, between about 0.5 Hertz to about 1.5 Hertz, between about 1 Hertz to about 2 Hertz, between about 1.5 Hertz to about 2.5 Hertz, between about 2 Hertz to about 3 Hertz, between about 2.5 Hertz to about 3.5 Hertz, between about 3 Hertz to about 4 Hertz, between about 3.5 Hertz to about 4.5 Hertz, between about 4 Hertz to about 5 Hertz, between about 4.5 Hertz to about 5.5 Hertz, between about 5 Hertz to about 6 Hertz, between about 5.5 Hertz to about 6.5 Hertz, between about 6 Hertz to about 7 Hertz, between about 6.5 Hertz to about 7.5 Hertz, between about 7 Hertz to about 8 Hertz, between about 7.5 Hertz to about 8.5 Hertz, between about 8 Hertz to about 9 Hertz, between about 8.5 Hertz to about 9.5 Hertz, and between about 9 Hertz to about 10 Hertz.

(168) The method of producing particular embodiments of the external ear canal pressure regulation device (1) can further include providing a third valve pressure relief element (72) having a configuration capable of coupling to the third valve (56), whereby the fluid flow generator (2) and the third valve pressure relief element (72) can be capable of alternate repeated operation to

generate the fluid flow (2) having the pressure wave (78) including the pressure wave amplitude (77) and the pressure wave frequency (79).

(169) As to particular embodiments, components of the external ear canal pressure regulation device (1) can be entirely formed of the same material, or alternatively, various components of the external ear canal pressure regulation device (1) can be formed from different materials.

Additionally, as to particular embodiments, the external ear canal pressure regulation device (1) or components of the external ear canal pressure regulation device (1) can be produced from any of a wide variety of processes depending upon the application, such as press molding, injection molding, fabrication, machining, printing, three-dimensional printing, or the like, or combinations thereof, as one piece or assembled from a plurality of pieces into an embodiment of the external ear canal pressure regulation device (1) or provided as a plurality of pieces for assembly into an embodiment of the external ear canal pressure regulation device (1).

(170) Components of the external ear canal pressure regulation device (1) can be produced from any of a wide variety of materials which can provide an embodiment of the external ear canal pressure regulation device (1) useful to generate and regulate a fluid flow (1). By way of non-limiting example, the valved conduit (1) can be produced from a variety of elastomeric compounds, plastic, plastic-like material, acrylic, polyamide, polyester, polypropylene, polyvinyl chloride-based materials, silicone-based materials, or the like, or combinations thereof.

(171) A method of using a particular embodiment of the external ear canal pressure regulation device (1) can include obtaining the external ear canal pressure regulation (1) device including a fluid flow generator (2) which generates a fluid flow; a valved conduit (5) fluidically coupled to the fluid flow generator (2), the valved conduit (5) having a first fluid flow conduit (49) interruptible by a first valve (52) to unidirectionally regulate the fluid flow (8) in the first fluid flow conduit (49); and an earpiece (3) having an axial earpiece bore (4) which communicates between an earpiece first end (31) and an earpiece second end (29), the axial earpiece bore (4) fluidically coupled to the valved conduit (5) opposite the fluid flow generator (2), the earpiece (3) having a compliant earpiece external surface (7) configured to sealably engage an external ear canal (6) as a barrier between an external ear canal pressure (10) and an ambient pressure (11); sealably engaging the earpiece external surface (7) of the earpiece (3) with the external ear canal (6); generating the fluid flow (8) between the fluid flow generator (2) and the axial earpiece bore (4); and regulating a pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11).

(172) As to particular embodiments, a method of using the external ear canal pressure regulation device (1) can include obtaining the external ear canal pressure regulation device (1) including a fluid flow generator (2) which generates a fluid flow (8); a valved conduit (5) having a first fluid flow conduit (49) interruptible by a first valve (52) to unidirectionally regulate the fluid flow (8) in the first fluid flow conduit (49); and an earpiece (3) having an axial earpiece bore (4) which communicates between an earpiece first end (31) and an earpiece second end (29), the earpiece (3) having a compliant earpiece external surface (7) configured to sealably engage an external ear canal (6) as a barrier between an external ear canal pressure (10) and an ambient pressure (11); fluidically coupling the valved conduit (5) in a first configuration (45) with the fluid flow generator (2) and the axial earpiece bore (4) of the earpiece (3) to unidirectionally regulate the fluid flow (8) in a first direction (47) in the first fluid flow conduit (49); sealably engaging the earpiece external surface (7) of the earpiece (3) with the external ear canal (6); generating the fluid flow (8) between the fluid flow generator (2) and the axial earpiece bore (4) in the first direction (47) in the first fluid flow conduit (49); and regulating a pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11) wherein the external ear canal pressure (10) is greater than the ambient pressure (11).

(173) As to particular embodiments, the method of using the external ear canal pressure regulation device (1) can further include operating a pressure relief element (70) to generate the fluid flow (8) from the external ear canal (6) toward the ambient pressure (11) to return the external ear canal

pressure (10) toward the ambient pressure (11). As to particular embodiments, the method can further include disengaging the earpiece external surface (7) of the earpiece (3) from the external ear canal (6).

(174) As to particular embodiments, the method of using the external ear canal pressure regulation device (1) can further include uncoupling the valved conduit (5) in the first configuration (45) from the fluid flow generator (2) and the axial earpiece bore (4) of the earpiece (3).

(175) As to particular embodiments, the method of using the external ear canal pressure regulation device (1) can further include fluidically coupling the valved conduit (5) in a second configuration (46) with the fluid flow generator (2) and the axial earpiece bore (4) of the earpiece (3) to unidirectionally regulate the fluid flow (8) in a second direction (48) in the first fluid flow conduit (49); sealably engaging the earpiece external surface (7) of the earpiece (3) with the external ear canal (7); generating the fluid flow (8) between the fluid flow generator (2) and the axial earpiece bore (4) in the second direction (48) in the first fluid flow conduit (49); and regulating a pressure differential (9) between the external ear canal pressure (10) and the ambient pressure (11) wherein the external ear canal pressure (10) is lesser than the ambient pressure (11).

(176) As to particular embodiments, the method of using the external ear canal pressure regulation device (1) can further include operating a pressure relief element (70) to generate the fluid flow (8) from the ambient pressure (11) toward the external ear canal (6) to return the external ear canal pressure (10) toward the ambient pressure (11). As to particular embodiments, the method can further include disengaging the earpiece external surface (7) of the earpiece (3) from the external ear canal (6).

(177) As to particular embodiments, the method of using the external ear canal pressure regulation device (1) can further include uncoupling the valved conduit (5) in the second configuration (46) from the fluid flow generator (2) and the axial earpiece bore (4) of the earpiece (3).

(178) As can be easily understood from the foregoing, the basic concepts of the present invention may be embodied in a variety of ways. The invention involves numerous and varied embodiments of an external ear canal pressure regulation device and methods for making and using such external ear canal pressure regulation devices including the best mode.

(179) As such, the particular embodiments or elements of the invention disclosed by the description or shown in the figures or tables accompanying this application are not intended to be limiting, but rather exemplary of the numerous and varied embodiments generically encompassed by the invention or equivalents encompassed with respect to any particular element thereof. In addition, the specific description of a single embodiment or element of the invention may not explicitly describe all embodiments or elements possible; many alternatives are implicitly disclosed by the description and figures.

(180) It should be understood that each element of an apparatus or each step of a method may be described by an apparatus term or method term. Such terms can be substituted where desired to make explicit the implicitly broad coverage to which this invention is entitled. As but one example, it should be understood that all steps of a method may be disclosed as an action, a means for taking that action, or as an element which causes that action. Similarly, each element of an apparatus may be disclosed as the physical element or the action which that physical element facilitates. As but one example, the disclosure of a “fluid flow” should be understood to encompass disclosure of the act of “flowing fluid”—whether explicitly discussed or not—and, conversely, were there effectively disclosure of the act of “flowing fluid”, such a disclosure should be understood to encompass disclosure of a “fluid flow” and even a “means for flowing fluid.” Such alternative terms for each element or step are to be understood to be explicitly included in the description.

(181) In addition, as to each term used it should be understood that unless its utilization in this application is inconsistent with such interpretation, common dictionary definitions should be understood to be included in the description for each term as contained in the Random House Webster's Unabridged Dictionary, second edition, each definition hereby incorporated by reference.

(182) All numeric values herein are assumed to be modified by the term “about”, whether or not explicitly indicated. For the purposes of the present invention, ranges may be expressed as from “about” one particular value to “about” another particular value. When such a range is expressed, another embodiment includes from the one particular value to the other particular value. The recitation of numerical ranges by endpoints includes all the numeric values subsumed within that range. A numerical range of one to five includes for example the numeric values 1, 1.5, 2, 2.75, 3, 3.80, 4, 5, and so forth. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint. When a value is expressed as an approximation by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. The term “about” generally refers to a range of numeric values that one of skill in the art would consider equivalent to the recited numeric value or having the same function or result. Similarly, the antecedent “substantially” means largely, but not wholly, the same form, manner or degree and the particular element will have a range of configurations as a person of ordinary skill in the art would consider as having the same function or result. When a particular element is expressed as an approximation by use of the antecedent “substantially,” it will be understood that the particular element forms another embodiment.

(183) Moreover, for the purposes of the present invention, the term “a” or “an” entity refers to one or more of that entity unless otherwise limited. As such, the terms “a” or “an”, “one or more” and “at least one” can be used interchangeably herein.

(184) Thus, the applicant(s) should be understood to claim at least: i) each of the external ear canal pressure regulation devices herein disclosed and described, ii) the related methods disclosed and described, iii) similar, equivalent, and even implicit variations of each of these devices and methods, iv) those alternative embodiments which accomplish each of the functions shown, disclosed, or described, v) those alternative designs and methods which accomplish each of the functions shown as are implicit to accomplish that which is disclosed and described, vi) each feature, component, and step shown as separate and independent inventions, vii) the applications enhanced by the various systems or components disclosed, viii) the resulting products produced by such systems or components, ix) methods and apparatuses substantially as described hereinbefore and with reference to any of the accompanying examples, x) the various combinations and permutations of each of the previous elements disclosed.

(185) The background section of this patent application provides a statement of the field of endeavor to which the invention pertains. This section may also incorporate or contain paraphrasing of certain United States patents, patent applications, publications, or subject matter of the claimed invention useful in relating information, problems, or concerns about the state of technology to which the invention is drawn toward. It is not intended that any United States patent, patent application, publication, statement or other information cited or incorporated herein be interpreted, construed or deemed to be admitted as prior art with respect to the invention.

(186) The claims set forth in this specification, if any, are hereby incorporated by reference as part of this description of the invention, and the applicant expressly reserves the right to use all of or a portion of such incorporated content of such claims as additional description to support any of or all of the claims or any element or component thereof, and the applicant further expressly reserves the right to move any portion of or all of the incorporated content of such claims or any element or component thereof from the description into the claims or vice-versa as necessary to define the matter for which protection is sought by this application or by any subsequent application or continuation, division, or continuation-in-part application thereof, or to obtain any benefit of, reduction in fees pursuant to, or to comply with the patent laws, rules, or regulations of any country or treaty, and such content incorporated by reference shall survive during the entire pendency of this application including any subsequent continuation, division, or continuation-in-part application thereof or any reissue or extension thereon.

(187) Additionally, the claims set forth in this specification, if any, are further intended to describe

the metes and bounds of a limited number of the preferred embodiments of the invention and are not to be construed as the broadest embodiment of the invention or a complete listing of embodiments of the invention that may be claimed. The applicant does not waive any right to develop further claims based upon the description set forth above as a part of any continuation, division, or continuation-in-part, or similar application.

Claims

1. A method for alleviating a user's headache pain, the method comprising: sealably engaging an earpiece of a fluid transfer device with an ear of the user as a barrier between an external ear canal pressure of an external ear canal of the user and an ambient pressure outside the external ear canal; and operating a fluid flow generator of the fluid transfer device to generate a fluid flow between the earpiece and the external ear canal, operating the fluid flow generator comprises at least one of: operating the fluid flow generator in a first configuration to increase the external ear canal pressure relative to the ambient pressure so as to generate a positive external ear canal pressure differential between the external ear canal pressure and the ambient pressure, or operating the fluid flow generator in a second configuration to decrease the external ear canal pressure relative to the ambient pressure so as to generate a negative external ear canal pressure differential between the external ear canal pressure and the ambient pressure, wherein at least one of the positive external ear canal pressure differential or the negative external ear canal pressure differential causes a physiological response that alleviates the headache pain.
2. The method of claim 1, wherein operating the fluid flow generator comprises operating the fluid flow generator in the first configuration to increase the external ear canal pressure relative to the ambient pressure so as to generate the positive external ear canal pressure differential between the external ear canal pressure and the ambient pressure.
3. The method of claim 2, wherein operating the fluid flow generator further comprises operating the fluid flow generator in the second configuration to decrease the external ear canal pressure relative to the ambient pressure so as to generate the negative external ear canal pressure differential between the external ear canal pressure and the ambient pressure.
4. The method of claim 1, wherein operating the fluid flow generator comprises operating the fluid flow generator in the second configuration to decrease the external ear canal pressure relative to the ambient pressure so as to generate the negative external ear canal pressure differential between the external ear canal pressure and the ambient pressure.
5. The method of claim 1, wherein the fluid flow generator comprises a valve.
6. The method of claim 5, wherein the fluid flow generator is in the first configuration when the valve is in a first position, and wherein the fluid flow generator is in the second configuration when the valve is in a second position.
7. The method of claim 5, wherein the valve comprises a unidirectional valve.
8. The method of claim 1, wherein at least one of the positive external ear canal pressure differential or the negative external ear canal pressure differential is configured to move a tympanic membrane of the ear.
9. The method of claim 1, wherein operating the fluid flow generator further comprises generating the fluid flow having a pressure wave.
10. The method of claim 9, wherein the pressure wave comprises a pressure wave amplitude and a pressure wave frequency.
11. A method for alleviating a user's headache pain, the method comprising: engaging a fluid transfer device with an ear of the user as a barrier between an external ear canal pressure of an external ear canal of the user and an ambient pressure outside the external ear canal, the fluid transfer device comprising: a fluid flow generator, and a valved conduit in fluid communication with the fluid flow generator; and operating the fluid transfer device to generate a fluid flow

between the fluid transfer device and the external ear canal, operating the fluid transfer device comprises at least one of: operating the fluid flow generator when the valved conduit is in a first configuration to increase the external ear canal pressure relative to the ambient pressure so as to generate a positive external ear canal pressure differential between the external ear canal pressure and the ambient pressure, or operating the fluid flow generator when the valved conduit is in a second configuration to decrease the external ear canal pressure relative to the ambient pressure so as to generate a negative external ear canal pressure differential between the external ear canal pressure and the ambient pressure, wherein at least one of the positive external ear canal pressure differential or the negative external ear canal pressure differential causes a physiological response that alleviates the headache pain.

12. The method of claim 11, wherein operating the fluid transfer device comprises operating the fluid flow generator when the valved conduit is in the first configuration to increase the external ear canal pressure relative to the ambient pressure so as to generate the positive external ear canal pressure differential between the external ear canal pressure and the ambient pressure.

13. The method of claim 12, wherein operating the fluid transfer device further comprises operating the fluid flow generator when the valved conduit is in the second configuration to decrease the external ear canal pressure relative to the ambient pressure so as to generate the negative external ear canal pressure differential between the external ear canal pressure and the ambient pressure.

14. The method of claim 11, wherein operating the fluid transfer device comprises operating the fluid flow generator when the valved conduit is in the second configuration to decrease the external ear canal pressure relative to the ambient pressure so as to generate the negative external ear canal pressure differential between the external ear canal pressure and the ambient pressure.

15. The method of claim 11, wherein the fluid transfer device comprises an earpiece.

16. The method of claim 15, wherein engaging the fluid transfer device with the ear comprises sealably engaging the earpiece with the external ear canal of the user.

17. The method of claim 11, wherein the headache pain comprises a migraine headache pain.

18. The method of claim 11, wherein at least one of the positive external ear canal pressure differential or the negative external ear canal pressure differential is configured to move a tympanic membrane of the user.

19. The method of claim 11, wherein operating the fluid transfer device further comprises generating the fluid flow having a pressure wave.

20. The method of claim 19, wherein the pressure wave comprises a pressure wave amplitude and a pressure wave frequency.
